

Lecture 4: atomic shells and classical models of chemical bonding-2

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

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Major points of last lecture

Chemical formulas, equations, and reaction yields

- 1. Empirical and molecular formulas**
- 2. Chemical formula and percentage composition**
- 3. Writing balanced chemical equations**
- 4. Mass relationships in chemical reactions**
- 5. Limiting reactant and percentage yield**

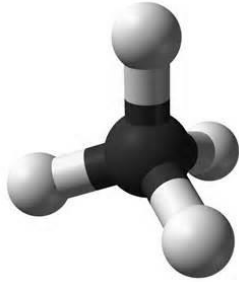
Empirical formula and the molecular formula

Molecular formula (分子式):

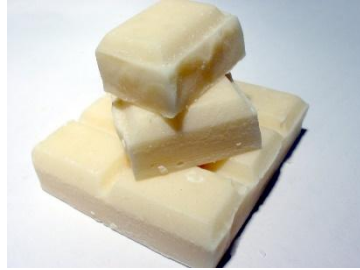
specifies the number of atoms of each element in one molecule of the substance.



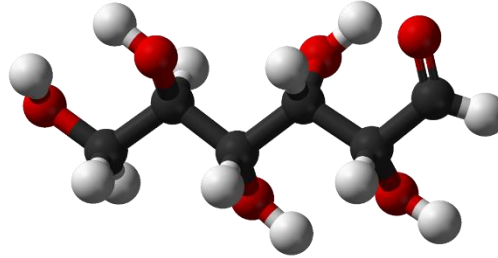
Methane



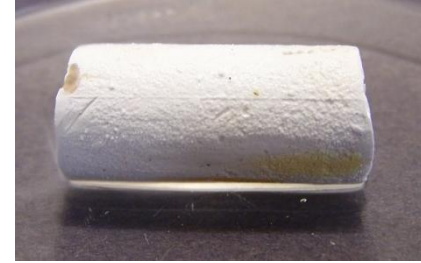
CH_4



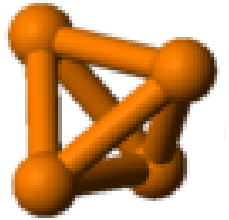
Glucose



$\text{C}_6\text{H}_{12}\text{O}_6$



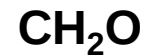
White phosphorus



P_4

Empirical formula (经验式):

the simplest formula that give the correct relative numbers of atoms of each kind in a compound.



Connection between empirical and molecular formulas

- The molecular formula is some whole-number multiple of the empirical formula.
- From Avogadro's hypothesis (same number of molecules of two gases in same volume, temperature and pressure), the ratio of molar masses of two gaseous compounds is the same as the ratio of their densities.

乙炔的摩尔质量

乙炔的密度

氧气的摩尔质量

$$\text{Molar mass of welding gas} = \frac{1.06 \text{ g L}^{-1}}{1.31 \text{ g L}^{-1}} (32.0 \text{ g mol}^{-1}) = 25.9 \text{ g mol}^{-1}$$

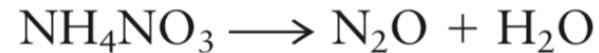
氧气的密度

乙炔= C_2H_2

Writing balanced chemical equations

Key: the **conservation of matter** in a chemical change is represented in a balanced chemical equation.

Stoichiometry (化学计量学): the study of the relationships between the numbers of reactant and product molecules.

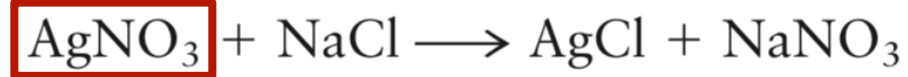


Steps:

- Assign 1 as the coefficient of one species. The best choice is the most complicated species; that is, the species with the largest number of elements.
- Identify, in sequence, elements that appear in only one chemical species, the coefficient of which has not yet been determined. Choose that coefficient to balance the number of moles of atoms of that element. Continue until all coefficients have been identified.
- If desired, multiply the whole equation by the smallest integer that will eliminate any fractions.

Limiting reactant and percentage yield

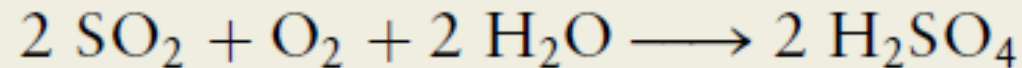
The one that is used up first: **limiting reactant**. These other reactants are present in **excess**.



limiting reactant

the silver nitrate is far more expensive than the sodium chloride

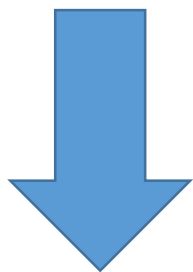
Sulfuric acid (H_2SO_4) forms in the chemical reaction



Suppose 400 g SO_2 , 175 g O_2 , and 125 g H_2O are mixed and the reaction proceeds until one of the reactants is used up. Which is the limiting reactant? What mass of H_2SO_4 is produced, and what masses of the other reactants remain?

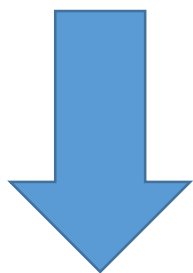
Percentage yield (收率)

Theoretical yields (理论产量) : the amounts of products calculated by assuming that the reaction goes cleanly and completely.



- Reaction stopped
- Competing reactions
- Lost during separation/purification

Actual yield (实际产量) < theoretical yield

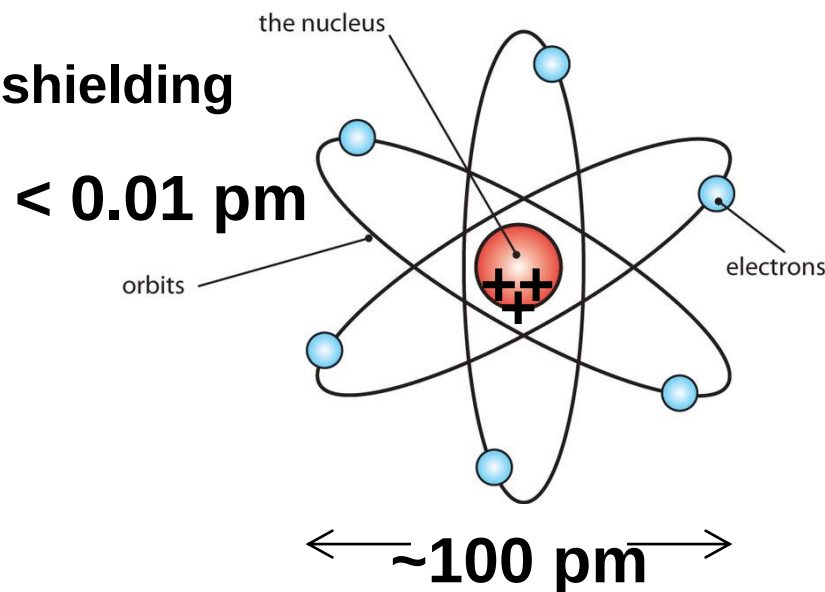


The ratio of the actual yield to the theoretical yield (multiplied by 100%) gives the percentage yield for that product in the reaction.

Major points of last lecture

- **Atomic shells and classical models of chemical bonding**

1. Representations of molecules
2. The periodic table
3. Forces and potential energy in atoms
4. Ionization energies, the shell model of the atom, and shielding



Molecular Formulas

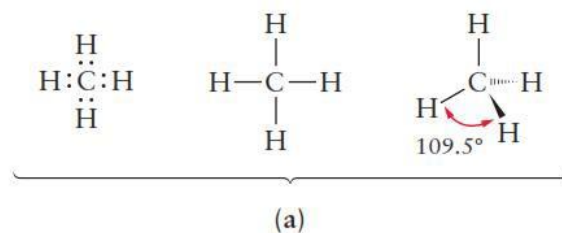
- ❑ Molecule: a collection of atoms bonded together, with the elements in fixed proportions and with a (generally) well-defined three-dimensional structure.

	Hydrogen	Water	Ammonia	Methane
Molecular formula	H_2	H_2O	NH_3	CH_4
Structural formula	$\text{H}-\text{H}$	$\text{H}-\text{O}-\text{H}$	$\begin{array}{c} \text{H}-\text{N}-\text{H} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$

- ❑ Held together by chemical forces.

Molecular representations

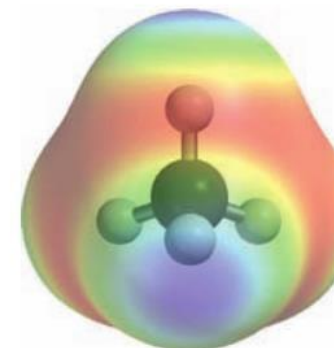
Methane



(b)

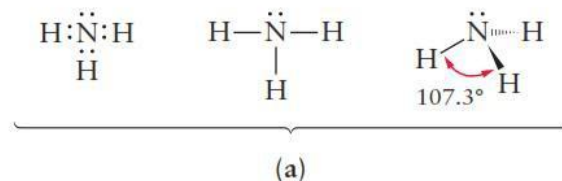


(c)

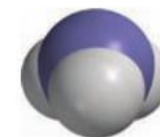


(d)

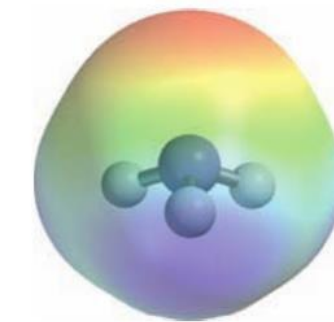
Ammonia



(b)

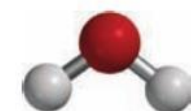
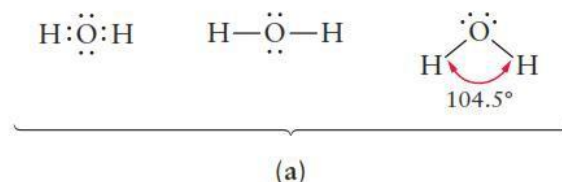


(c)



(d)

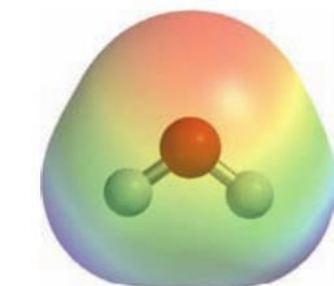
Water



(b)



(c)



(d)

Lewis dot

**line
structures**

**line
angle**

**ball and
stick**

**space
filling**

**electrostatic
potential
energy**

The Periodic Table

Representative elements

主族元素

Transition-metal elements

过渡金属元素

TRANSITION ELEMENTS

Groups (族): arranged vertically

Periods (周期): arranged horizontally

1 I	Transition-metal elements 过渡金属元素																18 VIII
1 H 1766	2 II	TRANSITION ELEMENTS										13 III	14 IV	15 V	16 VI	17 VII	2 He 1895
3 Li 1817	4 Be 1798											5 B 1808	6 C 1772	7 N 1772	8 O 1772	9 F 1887	10 Ne 1898
11 Na 1807	12 Mg 1756	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VIII	9 VIII	10 VIII	11 IB	12 IIB	13 Al 1827	14 Si 1823	15 P 1669	16 S 1774	17 Cl 1774	18 Ar 1894
19 K 1807	20 Ca 1808	21 Sc 1879	22 Ti 1791	23 V 1830	24 Cr 1797	25 Mn 1774	26 Fe 1735	27 Co 1751	28 Ni 1751	29 Cu 1746	30 Zn 1746	31 Ga 1875	32 Ge 1886	33 As 1817	34 Se 1817	35 Br 1826	36 Kr 1898
37 Rb 1861	38 Sr 1790	39 Y 1794	40 Zr 1789	41 Nb 1801	42 Mo 1778	43 Tc 1937	44 Ru 1844	45 Rh 1803	46 Pd 1803	47 Ag 1817	48 Cd 1817	49 In 1863	50 Sn 1863	51 Sb 1863	52 Te 1782	53 I 1811	54 Xe 1898
55 Cs 1860	56 Ba 1808	71 Lu 1907	72 Hf 1923	73 Ta 1802	74 W 1781	75 Re 1925	76 Os 1803	77 Ir 1803	78 Pt 1735	79 Au 1735	80 Hg 1735	81 Tl 1861	82 Pb 1861	83 Bi 1861	84 Po 1898	85 At 1940	86 Rn 1900
87 Fr 1939	88 Ra 1898	103 Lr 1961	104 Rf 1965	105 Db 1970	106 Sg 1976	107 Bh 1976	108 Hs 1984	109 Mt 1982	110 Uun 1994	111 Uuu 1995	112 Uub 1996		114 Uuq 1999				

LANTHANIDES

57 La 1839	58 Ce 1803	59 Pr 1885	60 Nd 1843	61 Pm 1947	62 Sm 1879	63 Eu 1896	64 Gd 1880	65 Tb 1843	66 Dy 1886	67 Ho 1879	68 Er 1843	69 Tm 1879	70 Yb 1907
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ACTINIDES

89 Ac 1899	90 Th 1828	91 Pa 1917	92 U 1789	93 Np 1940	94 Pu 1940	95 Am 1945	96 Cm 1944	97 Bk 1950	98 Cf 1950	99 Es 1952	100 Fm 1953	101 Md 1955	102 No 1958
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Lanthanide elements 镧系元素

Actinide elements 锕系元素

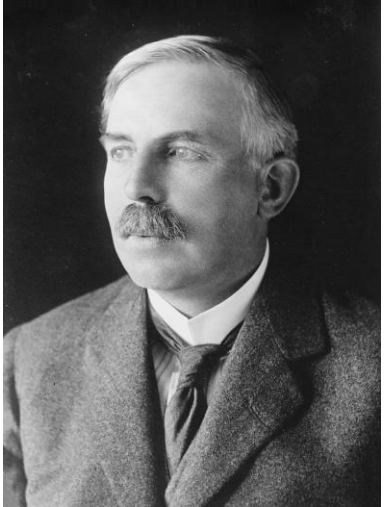
The Periodic Table

- Group I: alkali metals. Relatively soft metals with low melting points. LiCl and NaCl
- Group II: alkaline-earth metals. MgCl_2 and CaCl_2
- Group VI: chalcogens. CaO and BaS; Li_2O and Na_2S
- Group VII: halogens. LiF and LiCl

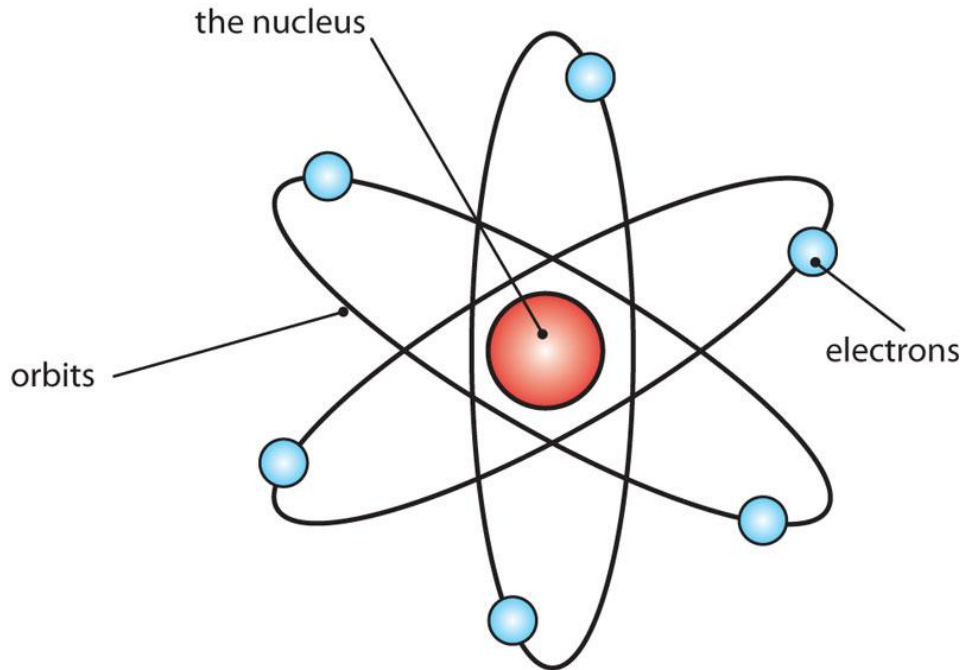
- Group VIII: **noble gases** (He, Ne, Ar, Kr, Xe, Rn), also called the **inert gases** because of their relative inertness toward chemical combination. (monatomic)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H Hydrogen 1.008	2 He Helium 4.0026																
3 Li Lithium 6.94	4 Be Beryllium 9.0122																
11 Na Sodium 22.990	12 Mg Magnesium 24.305																
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																	
57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97			
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (260)			

Forces and potential energy in atoms



Ernest Rutherford
(1871-1937)



Rutherford's planetary model of the atom assumes that an atom of atomic number Z comprises a dense, central nucleus of **positive charge $+Ze$** surrounded **by a total of Z electrons** moving around the nucleus.

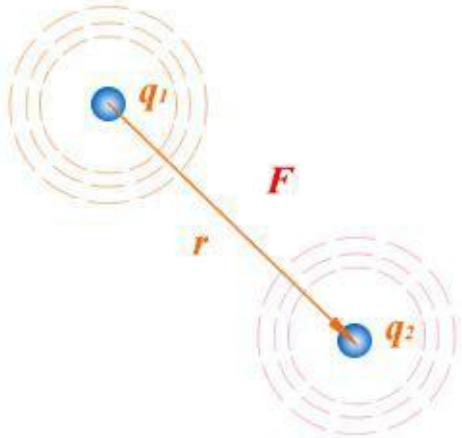
Structure and Behavior:

consequences of the **electrical forces** between the **electrons and the nucleus**.

Forces and potential energy in atoms

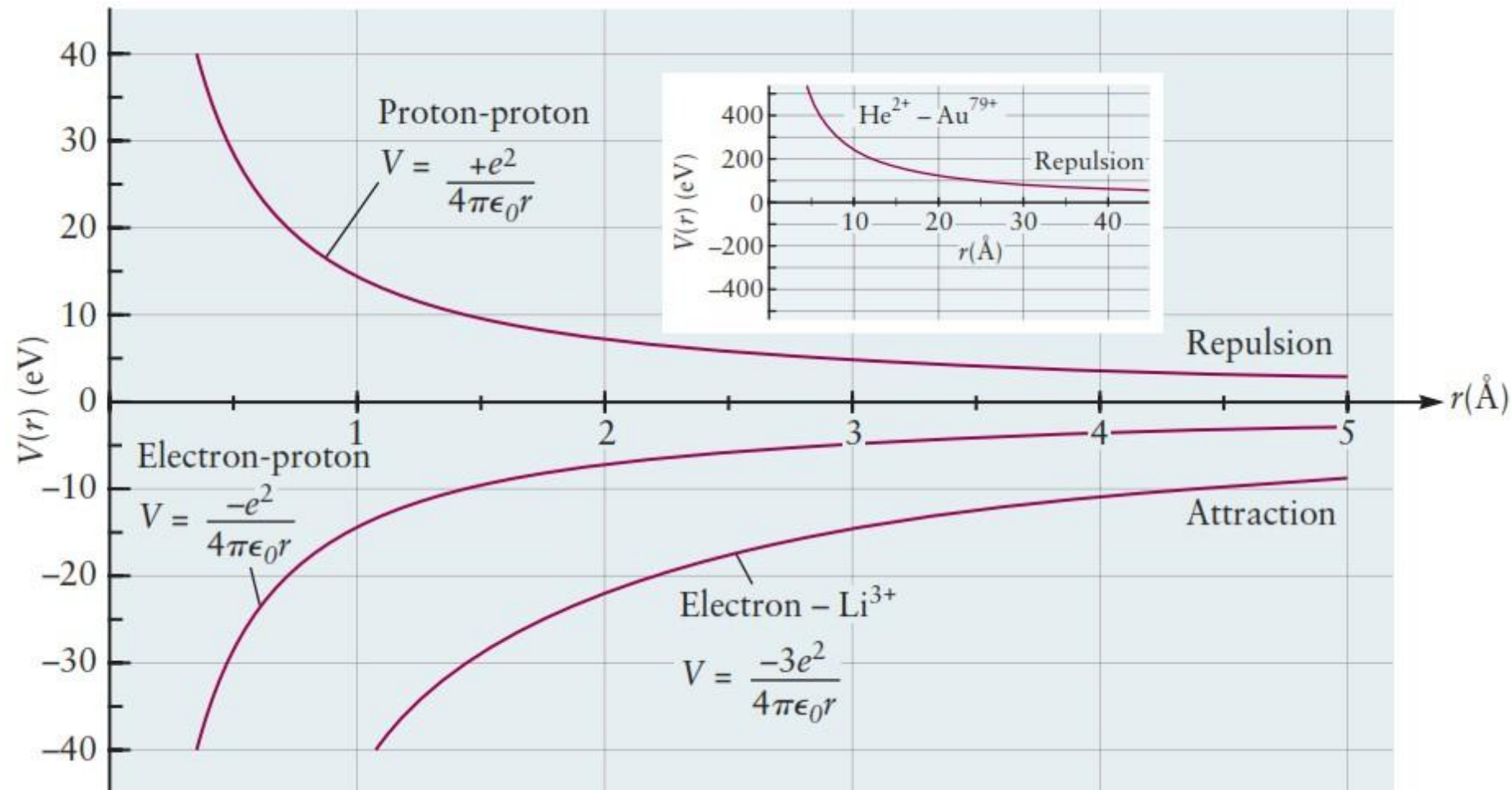
By convention, $V(r) \rightarrow 0$ as $r \rightarrow \infty$

$$F(r) = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$



$$V(r) = \frac{q_1 q_2}{4\pi\epsilon_0 r}$$

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$$



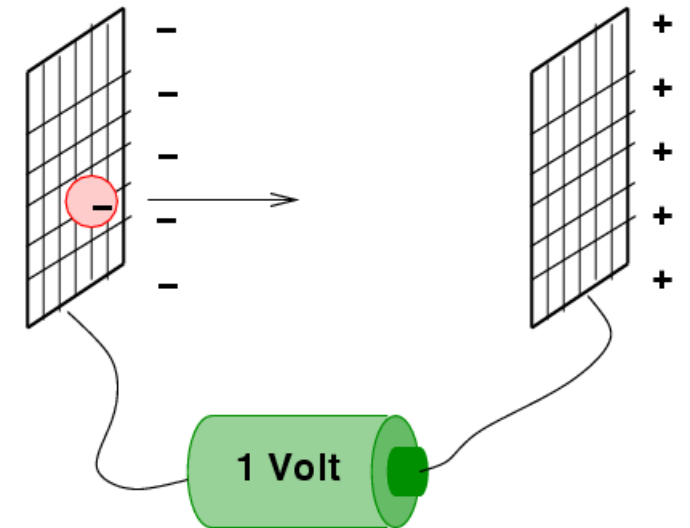
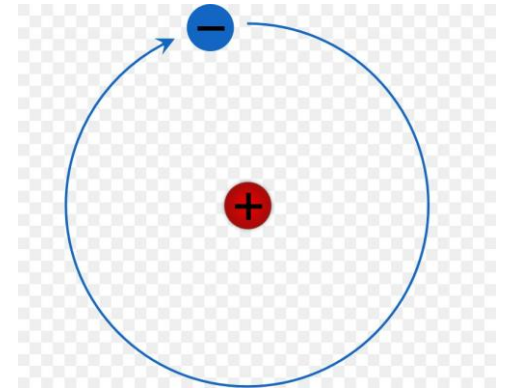
Forces and potential energy in atoms

- The typical **proton-electron distance** in a hydrogen atom is about 10^{-10} m.
- The **potential energy** of the hydrogen atom when the proton and electron are separated by a distance of $1 \text{ \AA}$ is

$$V(1\text{\AA}) = - \frac{(1.602 \times 10^{-19} \text{ C})^2}{4\pi(8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1})(1 \times 10^{-10} \text{ m})}$$
$$= - \frac{(8.988 \times 10^9)(1.602 \times 10^{-19})^2}{(1 \times 10^{-10})} \text{ J}$$

$$V(1\text{\AA}) = -2.307 \times 10^{-18} \text{ J}$$

$$V(1\text{\AA}) = - \frac{2.307 \times 10^{-18} \text{ J}}{1.602 \times 10^{-19} \text{ J(eV)}^{-1}} = -14.40 \text{ eV}$$

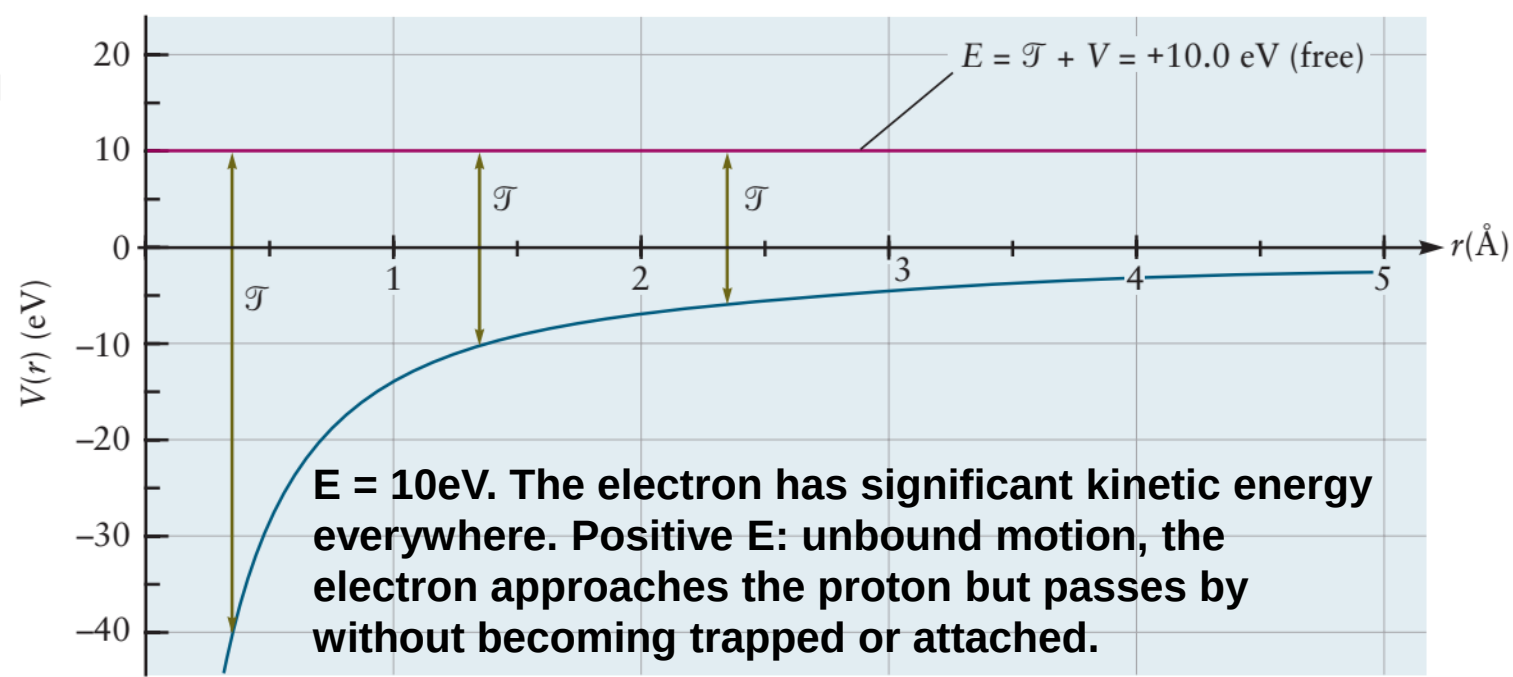


Electron volt 电子伏特

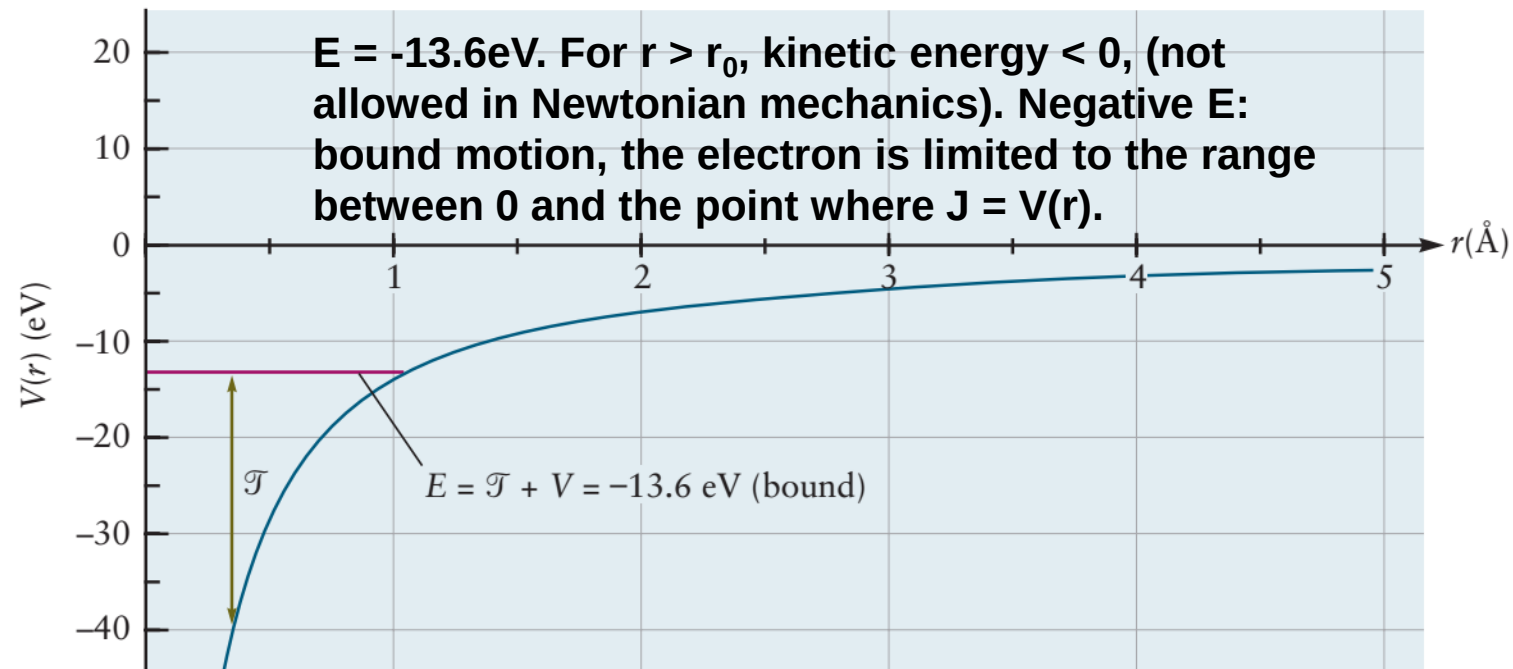
Total energy of electron in Hydrogen atom

The total energy (**kinetic + potential**) of the electron in the atom is

$$E = \frac{1}{2} m_e v^2 - \frac{Z e^2}{4 \pi \epsilon_0 r}$$

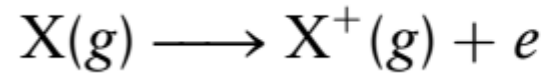


(a)



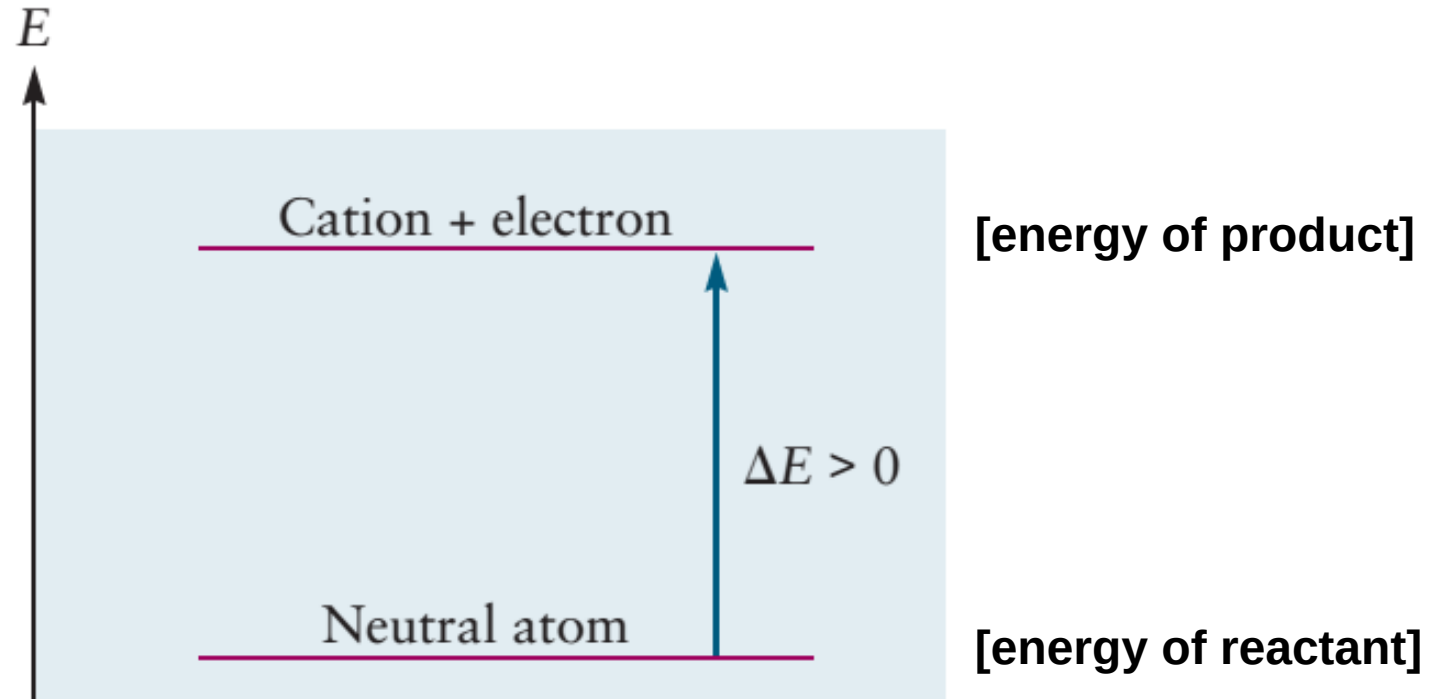
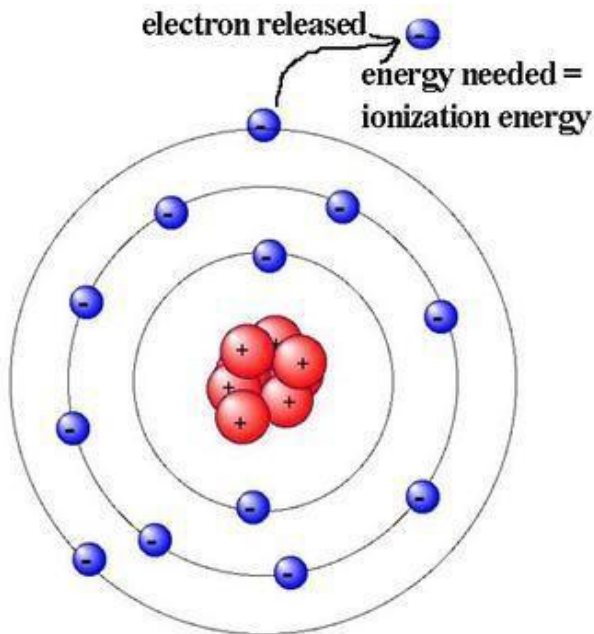
Ionization Energies (电离能)

The **ionization energy**, IE_1 , of an atom (first ionization energy) is the minimum energy necessary to detach an electron from the neutral gaseous atom and form a positively charged gaseous ion.



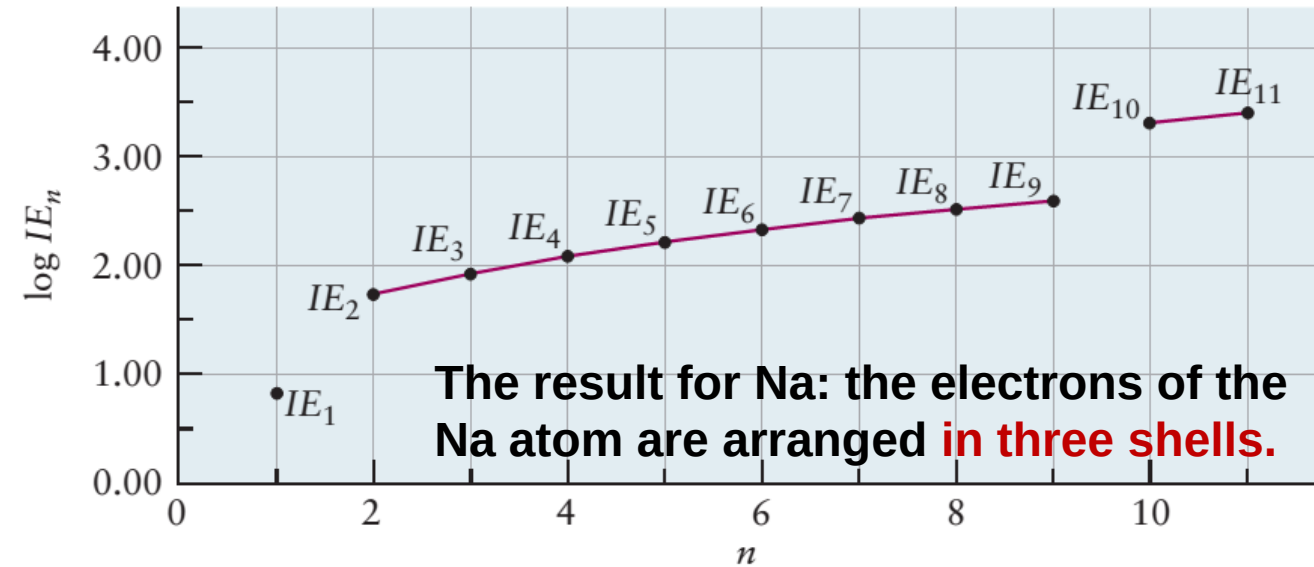
$$\Delta E = IE_1$$

$$\Delta E = [\text{energy of reaction products}] - [\text{energy of reactants}]$$

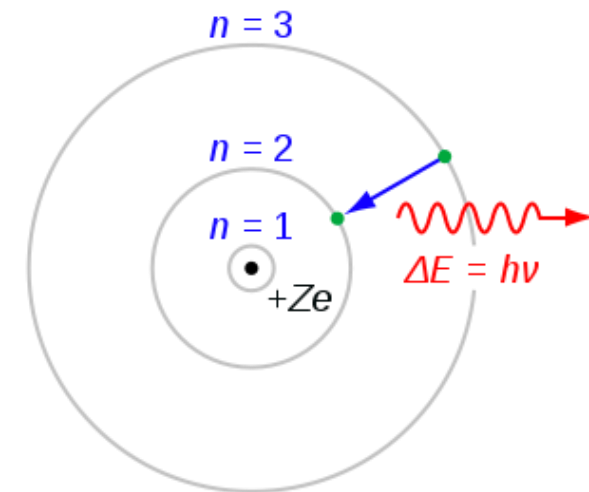


Ionization Energies (IE)

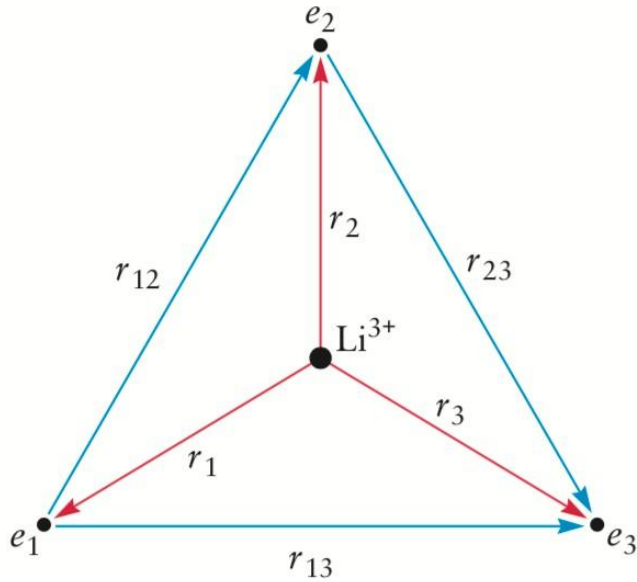
Z	Element	IE ₁	IE ₂	IE ₃	IE ₄	IE ₅	IE ₆	IE ₇	IE ₈	IE ₉	IE ₁₀
1	H	13.60									
2	He	24.59	54.42								
3	Li	5.39	75.64	122.45							
4	Be	9.32	18.21	153.89	217.71						
5	B	8.30	25.15	37.93	259.37	340.22					
6	C	11.26	24.38	47.89	64.49	392.08	489.99				
7	N	14.53	29.60	47.45	77.47	97.89	552.06	667.03			
8	O	13.62	35.12	54.93	77.41	113.90	138.12	739.32	871.39		
9	F	17.42	34.97	62.71	87.14	114.24	157.16	185.18	953.89	1103.08	
10	Ne	21.56	40.96	63.45	97.11	126.21	157.93	207.27	239.09	1195.79	1362.16
11	Na	5.14	47.29	71.64	98.91	138.39	172.15	208.47	264.18	299.87	1465.10
12	Mg	7.65	15.04	80.14	109.24	141.26	186.50	224.94	265.90	327.94	367.53
13	Al	5.99	18.83	28.45	119.99	153.71	190.47	241.43	284.59	330.21	398.57
14	Si	8.15	16.35	33.49	45.14	166.77	205.05	246.52	303.17	351.10	401.43
15	P	10.49	19.73	30.18	51.37	65.02	220.43	263.22	309.41	371.73	424.50
16	S	10.36	23.33	34.83	47.30	72.68	88.05	280.93	328.23	379.10	447.10
17	Cl	12.97	23.81	39.61	53.46	67.8	97.03	114.19	348.28	400.03	455.62
18	Ar	15.76	27.63	40.74	59.81	75.02	91.01	124.32	143.46	422.43	478.68
19	K	4.34	31.63	45.72	60.91	82.66	100.0	117.56	154.86	175.82	503.44
20	Ca	6.11	11.87	50.91	67.10	84.41	108.78	127.7	147.24	188.54	211.27
21	Sc	6.54	12.80	24.76	73.47	91.66	111.1	138.0	158.7	180.02	225.32



- The **first electron** is easily removed to produce Na⁺, which has **neon like stability** as indicated by the large jump between IE₁ and IE₂.
- **Electrons 2 through 9** occupy the **second shell**, and all of them are more tightly bound than the first electron.
- A big jump between IE₉ and IE₁₀ suggests that the last two electrons occupy a **third shell**, the electrons of which are the most tightly bound of all.



From one electron atom to multi-electron atom



Li atom with three electrons.

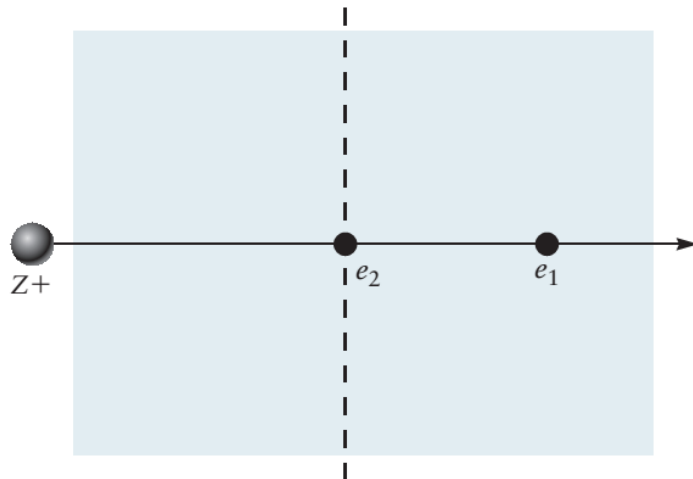
$$V = \frac{e^2}{4\pi\epsilon_0} \left(-\frac{Z}{r_1} - \frac{Z}{r_2} - \frac{Z}{r_3} + \frac{1}{r_{12}} + \frac{1}{r_{13}} + \frac{1}{r_{23}} \right)$$

Potential energy

How to simplify?



Effective potential energy = attractive nuclear-electron force + repulsive electron-electron force

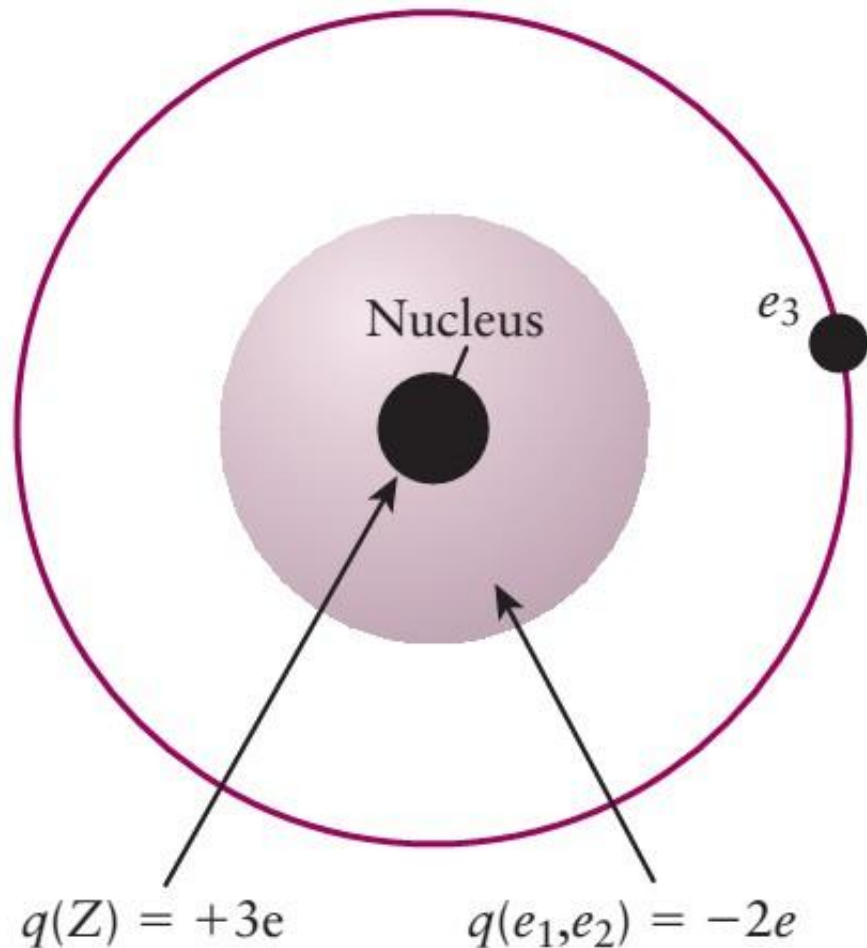


$$Z_{\text{eff}} = Z - S \quad \text{有效原子序数}$$

$$V_{\text{eff}}(r) = -\frac{Z_{\text{eff}}e^2}{4\pi\epsilon_0 r}$$

The shell Model of the atom

Lewis (路易斯) dot symbol



- The valence electrons as dots arranged around the chemical symbol
- The core electrons are not shown.
- The first four dots are arranged individually around the four sides of the symbol.
- If an atom has more than four valence electrons, dots representing them are then paired with those already present.

·H

·He

·Li

·Be·

·B·

·C·

·N·

·O·

·F·

·Ne·

Major points of today's lecture

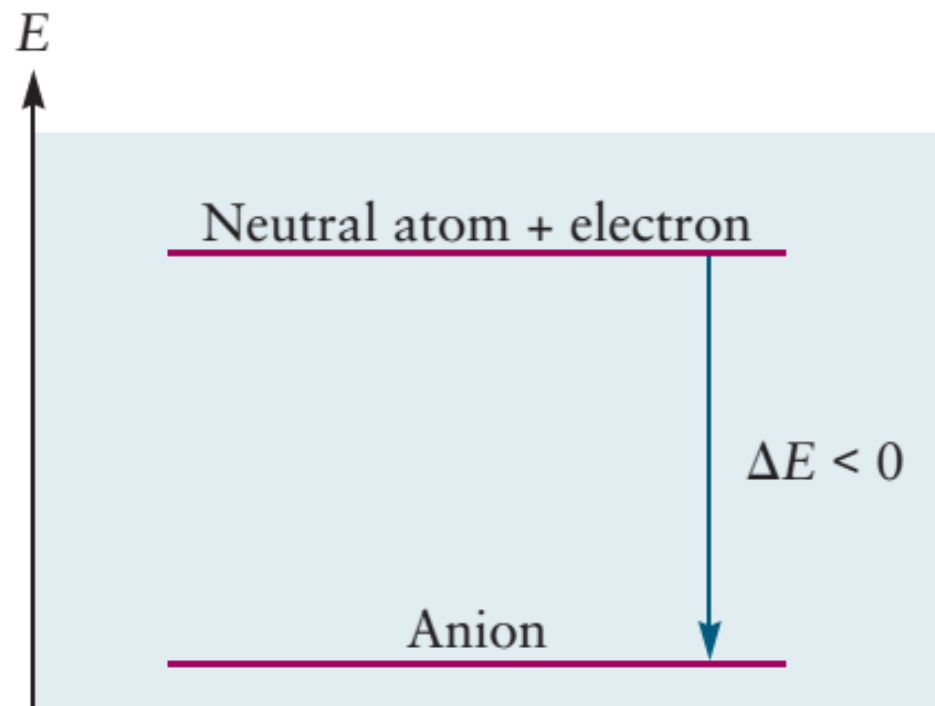
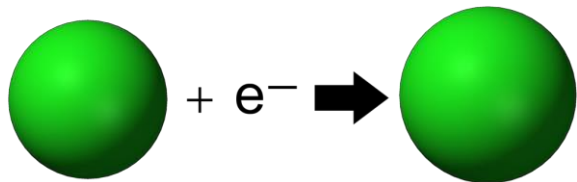
- **Atomic shells and classical models of chemical bonding**

1. **Electron affinity**

2. **Electronegativity: the tendency of atoms to attract electrons in molecules**

Electron affinity

Electron attachment energy (电子附着能): an atom accepts an extra electron to form an anion.

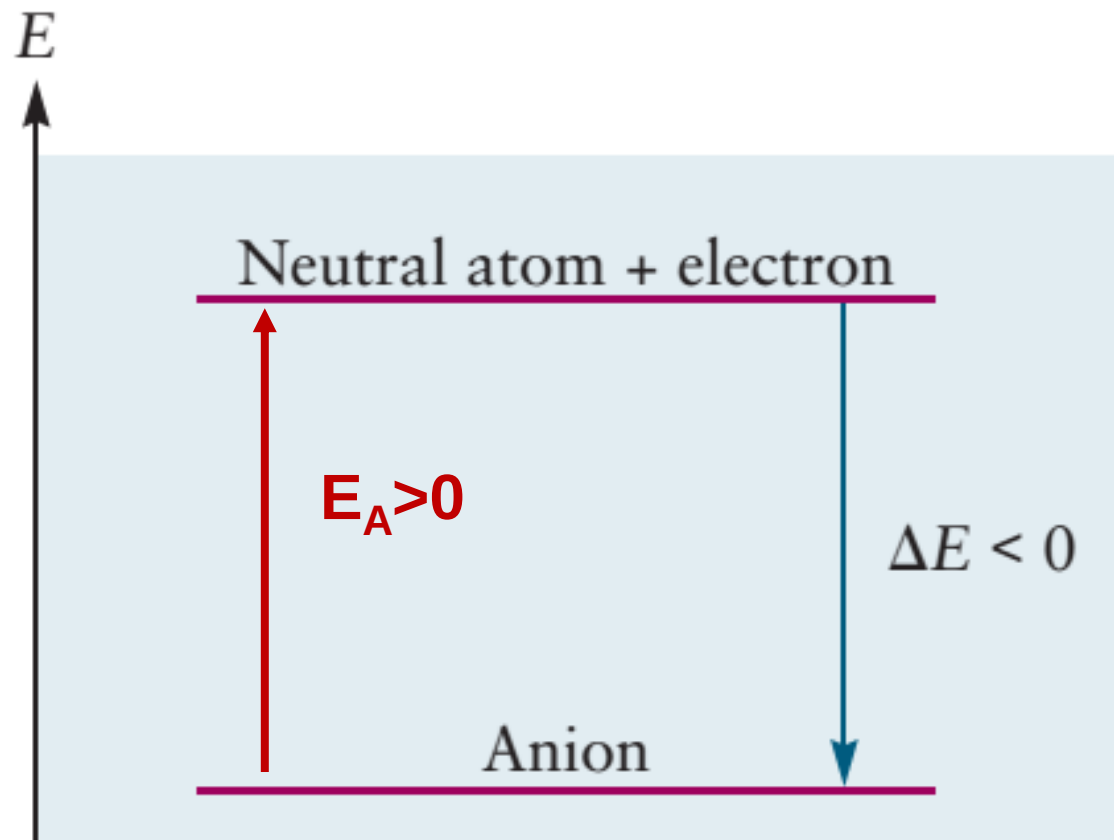
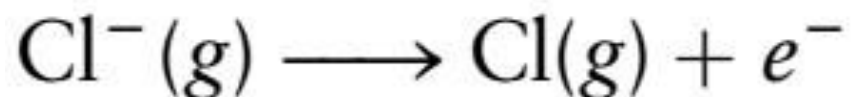


Usually, $\Delta E < 0$

Difficult to measure electron attachment energy!

Electron affinity

Electron affinity (EA_X) (电子亲合能) of atom X: energy required to detached the electron from the anion X^- and give the neutral atom.

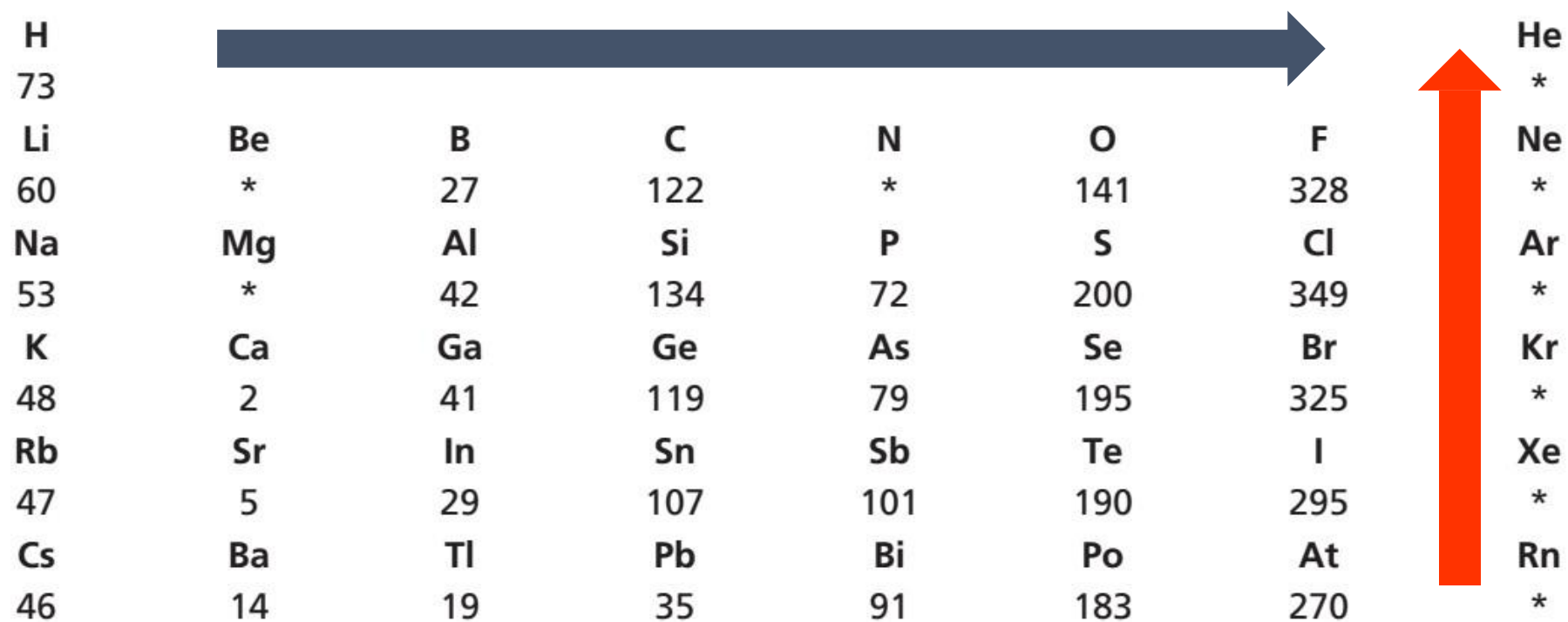


$$EA_{Cl} = 349 \text{ kJ/mol}$$

Electron affinity

TABLE 3.2

Electron Affinity of Selected Atoms (in kJ mol^{-1})



H							He
73							*
Li	Be	B	C	N	O	F	Ne
60	*	27	122	*	141	328	*
Na	Mg	Al	Si	P	S	Cl	Ar
53	*	42	134	72	200	349	*
K	Ca	Ga	Ge	As	Se	Br	Kr
48	2	41	119	79	195	325	*
Rb	Sr	In	Sn	Sb	Te	I	Xe
47	5	29	107	101	190	295	*
Cs	Ba	Tl	Pb	Bi	Po	At	Rn
46	14	19	35	91	183	270	*

*No stable anion A^- exists for this element in the gas phase.

No gaseous atom has a positive electron affinity for a second electron.

Electron affinity comparison

19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
---------	----------	----------	----------	---------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Higher Z_{eff}
Same distance



Higher electron
ionization energy

Higher Z_{eff}
Same distance



Ion is more
stable



Higher electron
affinity

Consider the elements selenium (Se) and bromine (Br). Without consulting tables of data, predict which one should have the higher value of ionization energy IE_1 . Predict which should have the higher value of electron affinity EA .

Solution

These elements are in adjacent groups in the 4th period of the periodic table. Progressing from selenium to bromine adds one unit of positive charge, which increases Z_{eff} . At the same time we add one more electron into the 4th shell like the valence electrons in Se. Since the nuclear attraction is greater for the added electron in bromine and its distance from the nucleus is about the same, it will be more tightly bound than the valence electrons in selenium. Therefore bromine will have the larger value of IE_1 .

If we place an additional electron in the 4th shell of each atom to form an anion, the additional electron on bromine will experience a larger value of Z_{eff} , and the Br^- anion will have greater stability than the Se^- anion. Moreover, Br^- has a stable octet configuration whereas Se^- does not. So Br has the larger value of EA .

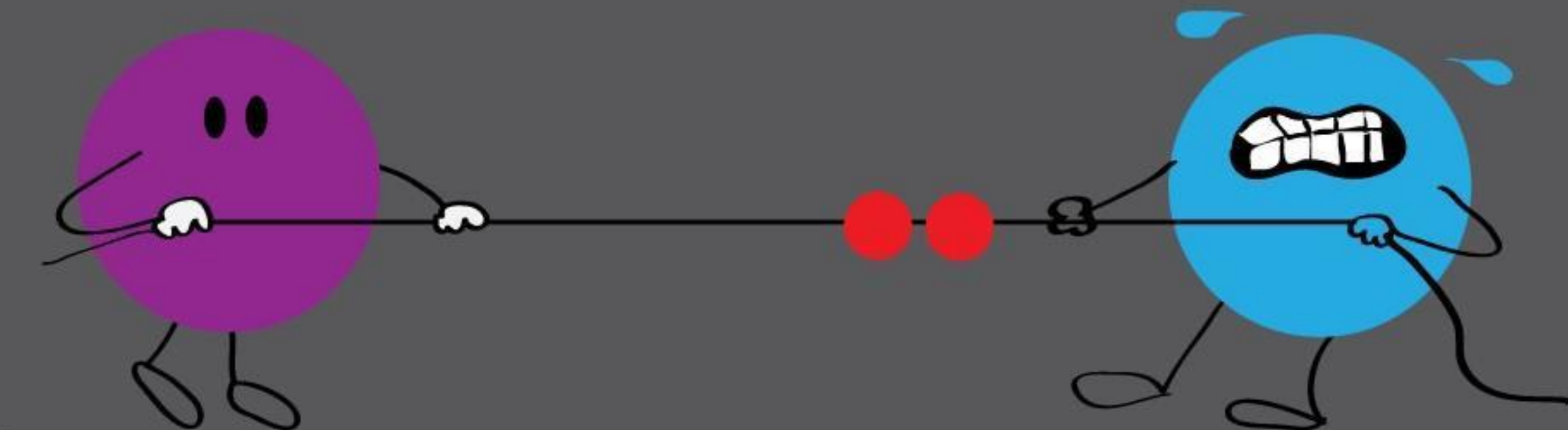
Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**
 1. Representations of molecules
 2. The periodic table
 3. Forces and potential energy in atoms
 4. Ionization energies, the shell model of the atom, and shielding
 5. Electron affinity
 6. Electronegativity: the tendency of atoms to attract electrons in molecules

ATOM A
weak pull

BIG
electronegativity
difference

ATOM B
strong pull



Can we define it ?

Electronegativity

Electronegativity (电负性): the tendency of atoms to attract electrons in molecules.

Mulliken's Electronegativity

$$\text{EN (Mulliken)} \propto \frac{1}{2} (IE_1 + EA)$$



Robert C. Mulliken
(1896–1986)

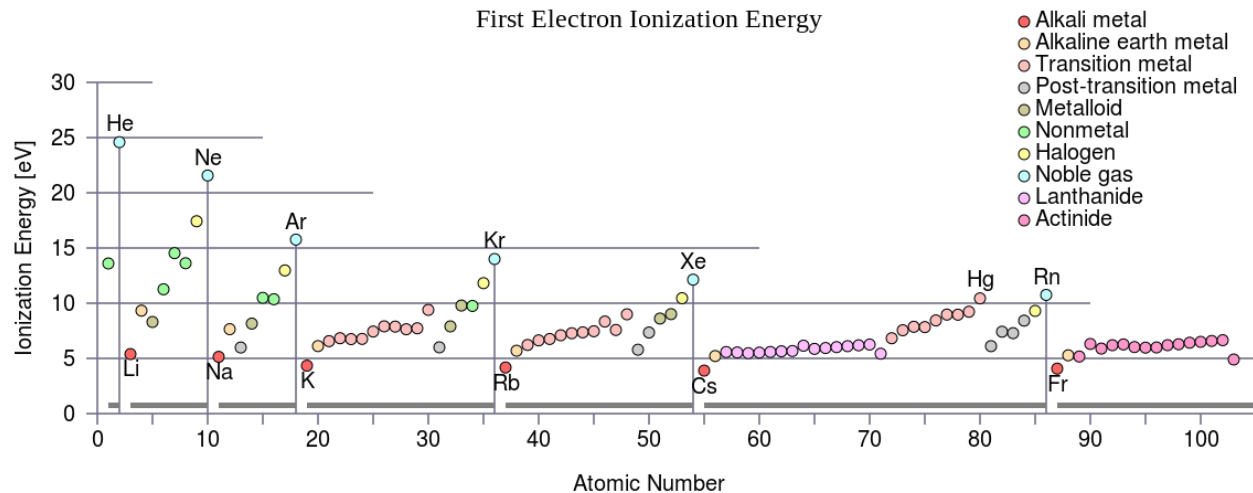
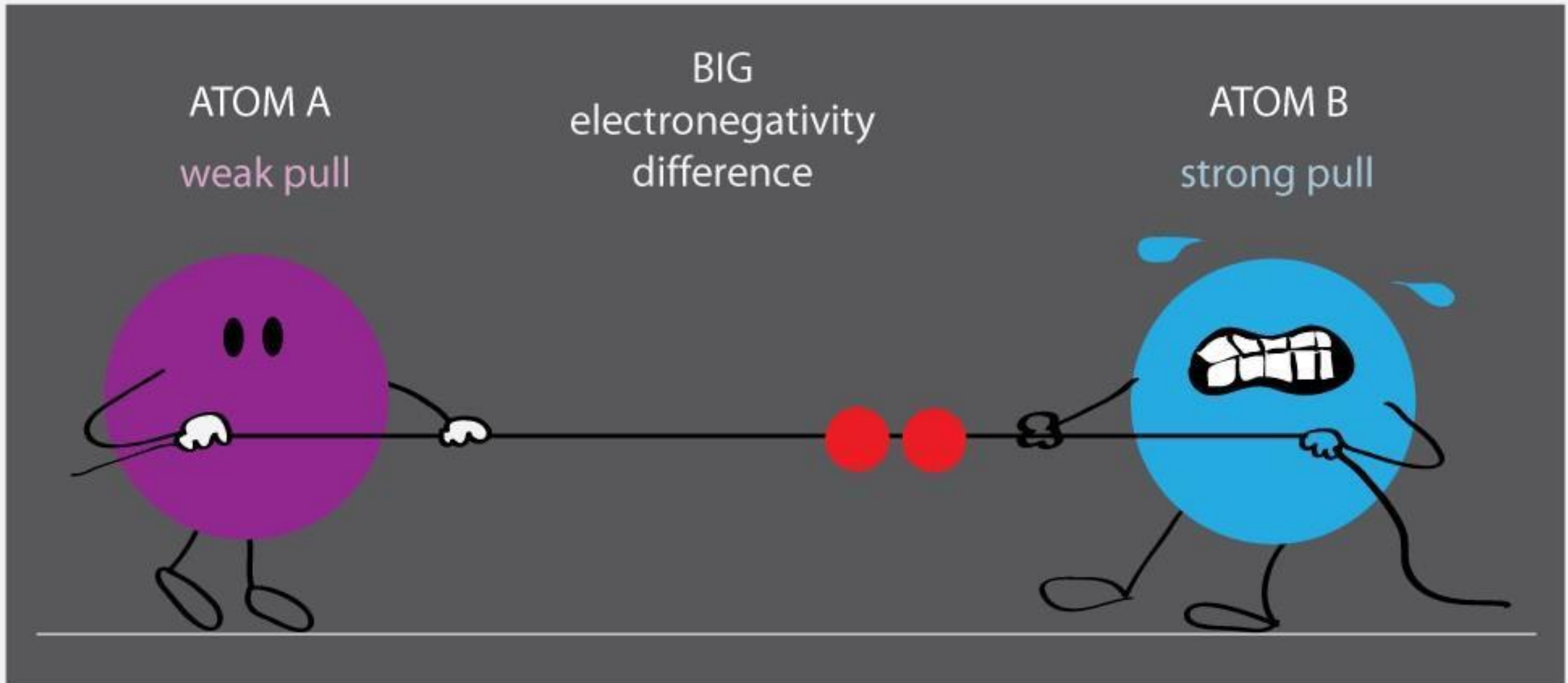


TABLE 3.2

Electron Affinity of Selected Atoms (in kJ mol⁻¹)

H							He
73							*
Li	Be	B	C	N	O	F	Ne
60	*	27	122	*	141	328	*
Na	Mg	Al	Si	P	S	Cl	Ar
53	*	42	134	72	200	349	*
K	Ca	Ga	Ge	As	Se	Br	Kr
48	2	41	119	79	195	325	*
Rb	Sr	In	Sn	Sb	Te	I	Xe
47	5	29	107	101	190	295	*
Cs	Ba	Tl	Pb	Bi	Po	At	Rn
46	14	19	35	91	183	270	*

*No stable anion A⁻ exists for this element in the gas phase.



- **Electron donors** have small ionization energies and small electron affinities (**low electronegativities**); they are highly **electropositive**.
- **Electron acceptors** have large ionization energies and large electron affinities (**high electronegativities**); they are highly **electronegative**.

The Nobel Prize in Chemistry 1966

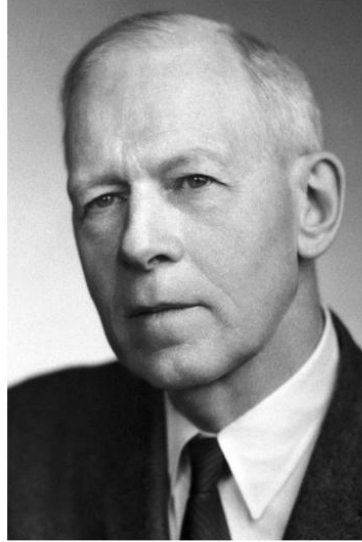
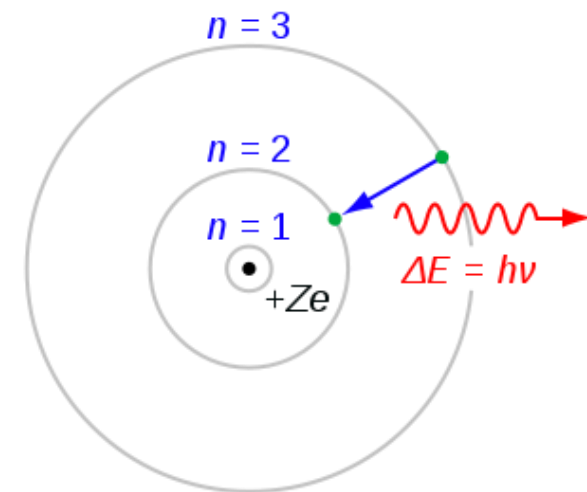
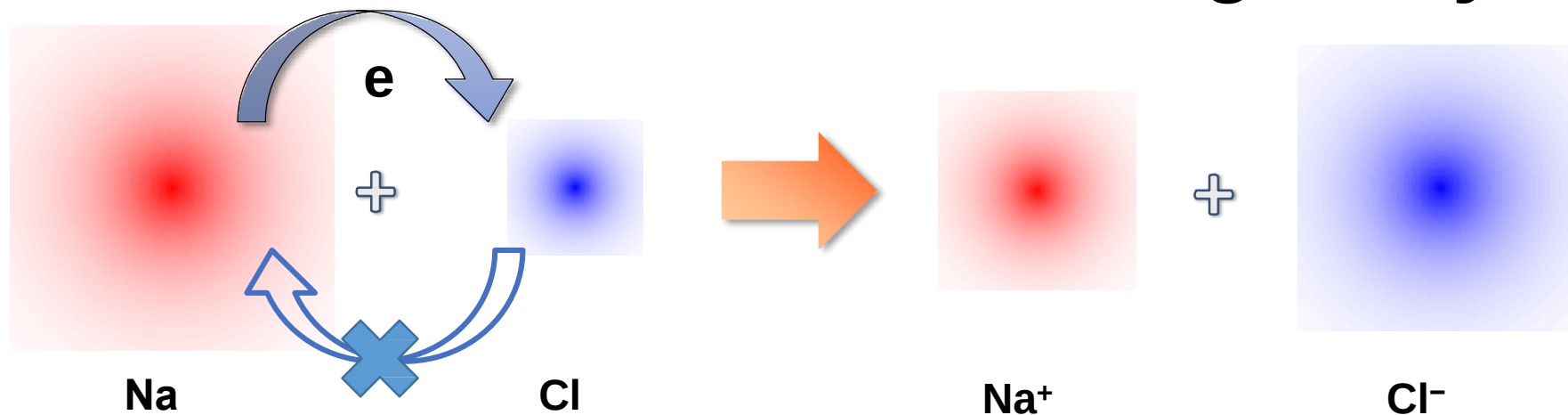


Photo from the Nobel
Foundation archive.
Robert S. Mulliken

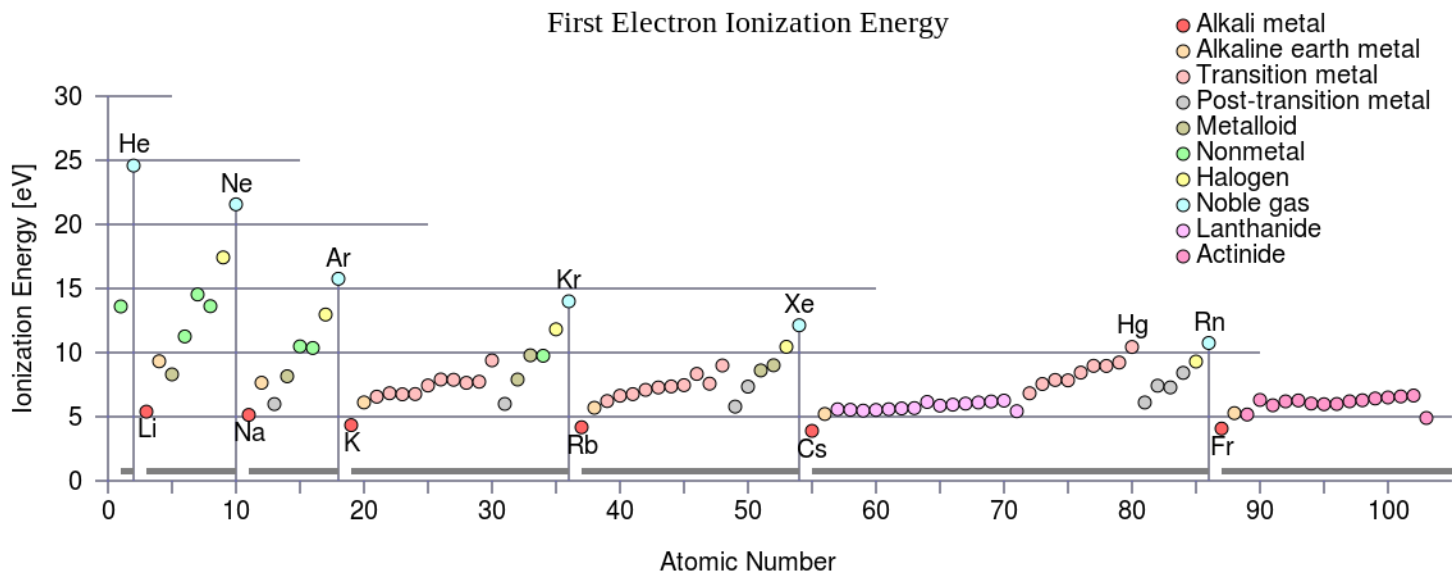
The Nobel Prize in Chemistry 1966 was awarded to was awarded to Robert S. Mulliken
"for his fundamental work concerning chemical bonds and the electronic structure of
molecules by the molecular orbital method"

<https://www.nobelprize.org/prizes/chemistry/1966/summary/>

Mulliken's Electronegativity



First Electron Ionization Energy



T A B L E 3.2

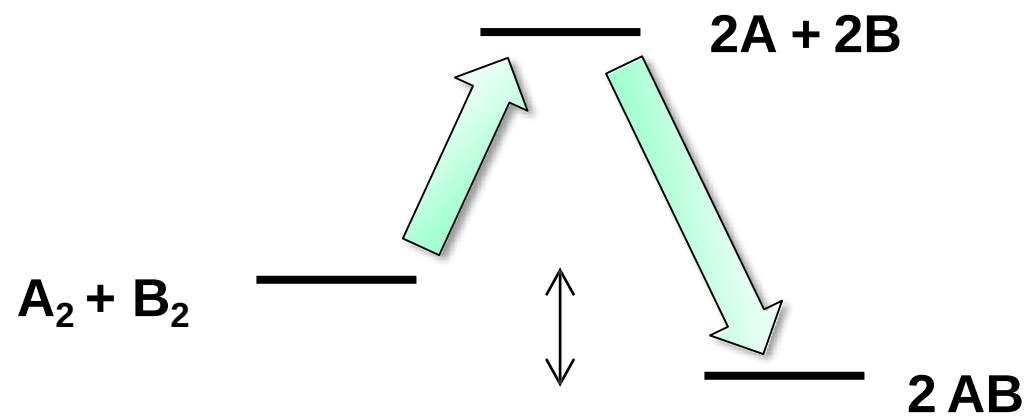
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48	2	41	119	79	195	325	*
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46	14	19	35	91	183	270	*

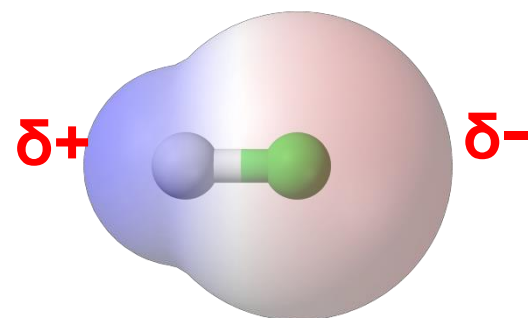
*No stable anion A⁻ exists for this element in the gas phase.

Pauling's Electronegativity

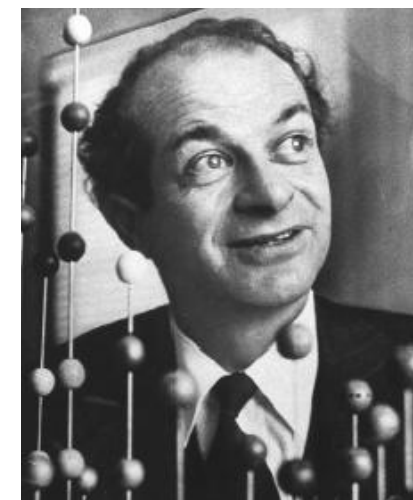
- NaCl vs. HF, HCl, HBr vs. H₂, F₂, Cl₂, Br₂



$$2 \times E_{\text{bond}} (\text{H-Cl}) > E_{\text{bond}} (\text{H-H}) + E_{\text{bond}} (\text{Cl-Cl})$$



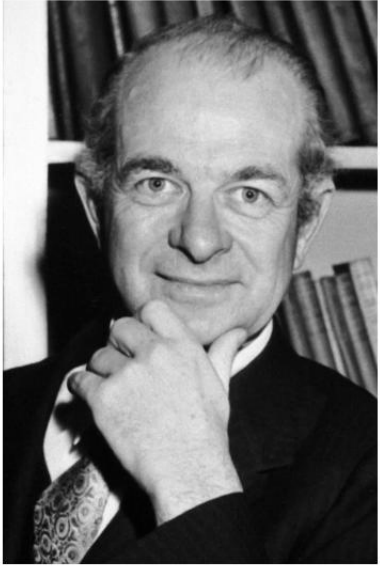
$$\Delta \propto q^2$$



Linus Pauling
(1901–1994)

The extra stability was provided by partial **charge separation** in the bond, called **ionic character**.

The Nobel Prize in Chemistry 1954



Linus Carl Pauling

Prize share: 1/1

The Nobel Prize in Chemistry 1954 was awarded to Linus Carl Pauling "for his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances"

<https://www.nobelprize.org/prizes/chemistry/1954/summary/>
<https://www.nobelprize.org/prizes/peace/1962/summary/>

The Nobel Peace Prize 1962

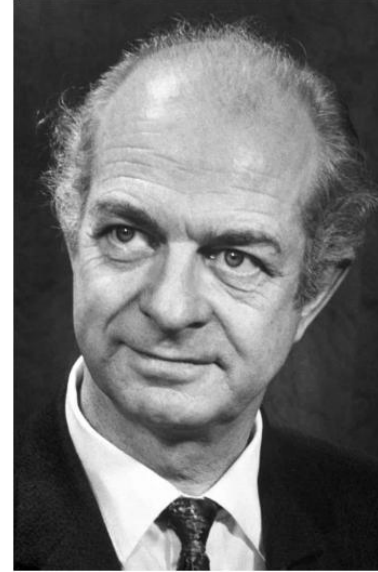


Photo from the Nobel
Foundation archive.

Linus Carl Pauling

Prize share: 1/1

The Nobel Prize in Peace 1962 was awarded to Linus Carl Pauling "for his fight against the nuclear arms race between East and West"

Pauling's Electronegativity

Pauling's Electronegativity scale: a **comparison of the bond energies** of a large number of **heteronuclear bond pairs** with those of **homonuclear diatomics** comprising the same elements.

An empirical formula:

1. Dissociation energy of A-A bond and B-B bond (both are covalent bonds): ΔE_{AA} , ΔE_{BB}
2. Dissociation energy of A-B bond (only **covalent contribution part**): $(\Delta E_{AA} \times \Delta E_{BB})^{0.5}$
3. Dissociation energy of A-B bond (**covalent contribution + ionic contribution**): ΔE_{AB}

Excess bond energy $\Delta = \Delta E_{AB} - \sqrt{\Delta E_{AA} \Delta E_{BB}}$

Pauling's definition of **electronegativity difference**

$$\chi_A - \chi_B = 0.102 \Delta^{1/2}$$

Pauling's Electronegativity

Pauling arbitrarily assigned the value 4.0 to F.

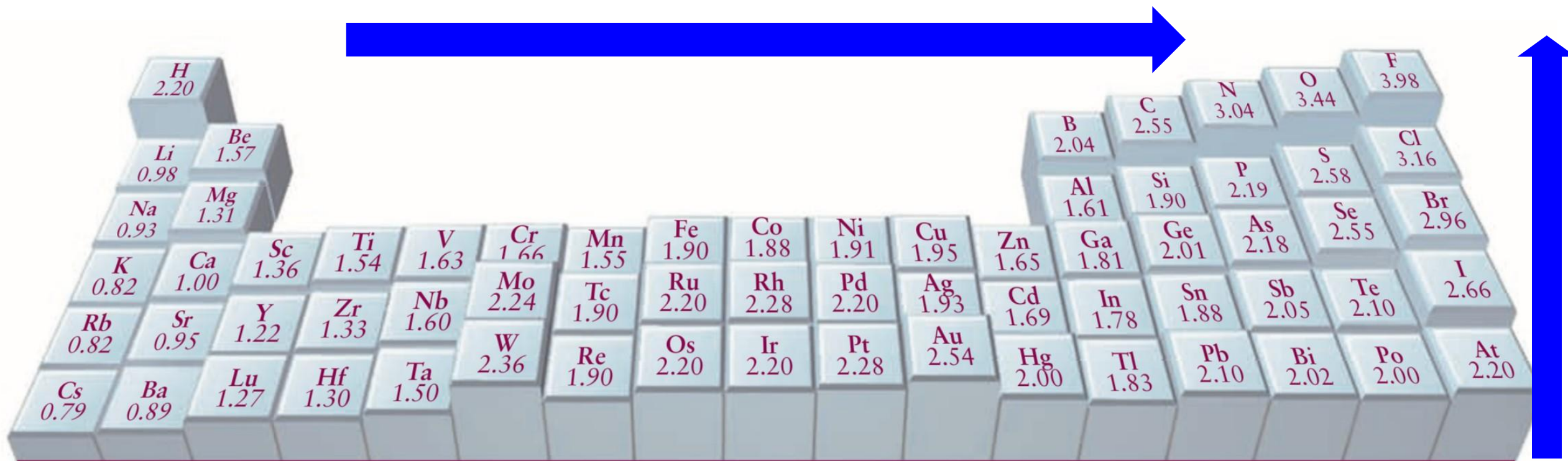


FIGURE 3.7 Average electronegativity of atoms, computed with the method that Linus Pauling developed. Electronegativity values have no units.

Without consulting tabulated values of electronegativity and guided only by the periodic table, arrange these atoms in order of increasing electronegativity: H, F, Al, O. Give a brief explanation of your answer.

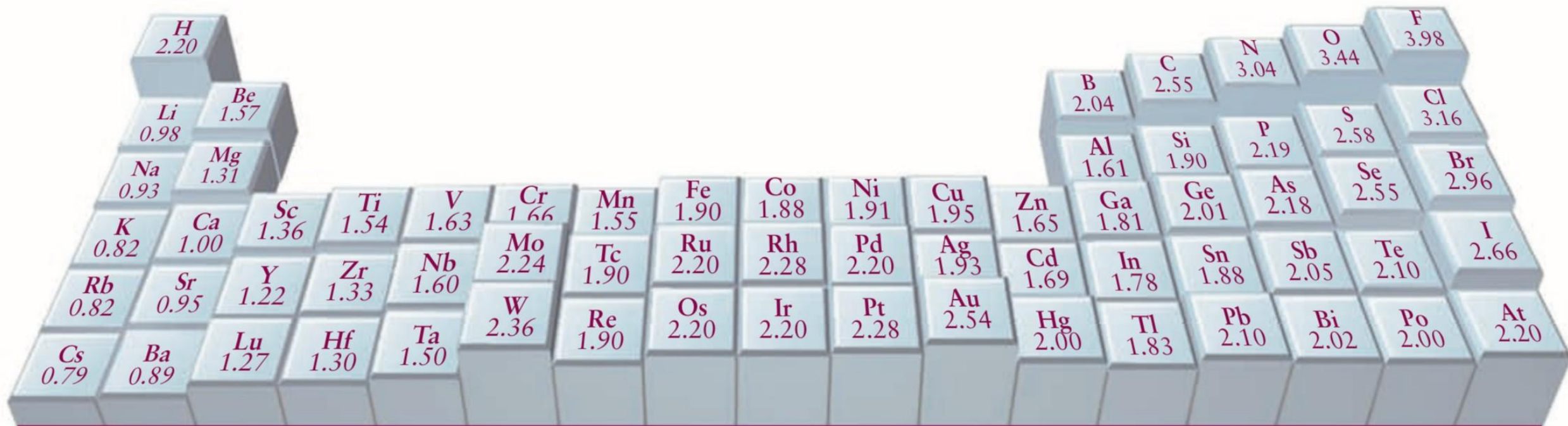


FIGURE 3.7 Average electronegativity of atoms, computed with the method that Linus Pauling developed. Electronegativity values have no units.

Major points of today's lecture

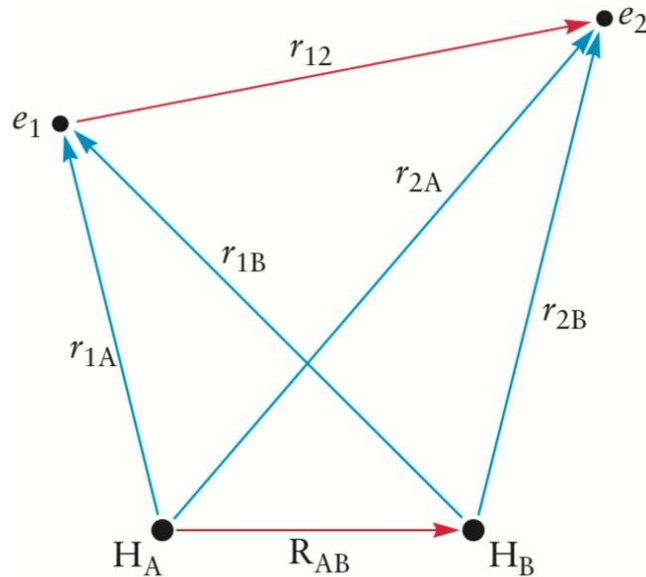
- **Atomic shells and classical models of chemical bonding**
 1. **Forces and potential energy in molecules: formation of chemical bonds**
 2. **Ionic bonding**
 3. **Covalent and polar covalent bonding**
 4. **Electron pair bonds and Lewis diagrams for molecules**
 5. **The shapes of molecules**
 6. **Oxidation numbers**
 7. **Inorganic nomenclature**

Forces and potential energy in molecules

Stability of a chemical bond? Why is the H₂ molecule more stable than a pair of separated hydrogen atoms in the gas phase at normal temperatures and pressures?



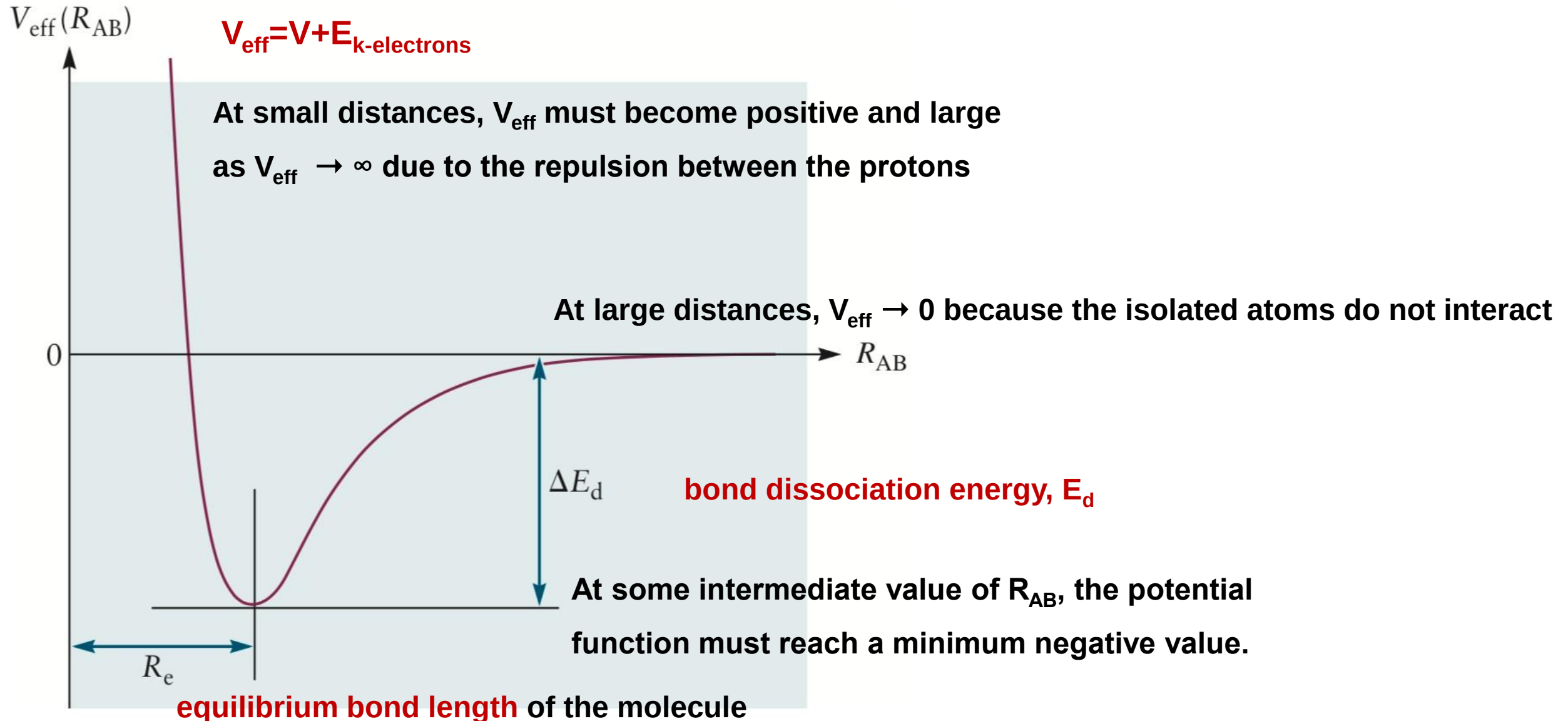
total energy of H₂ molecule < total energy of two separated H atoms.



$$V = -\frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r_{1A}} + \frac{1}{r_{2A}} + \frac{1}{r_{1B}} + \frac{1}{r_{2B}} \right) + \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r_{12}} \right) + \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{R_{AB}} \right)$$

$$V = V_{en} + V_{ee} + V_{nn}$$

Forces and potential energy in molecules



Major points of today's lecture

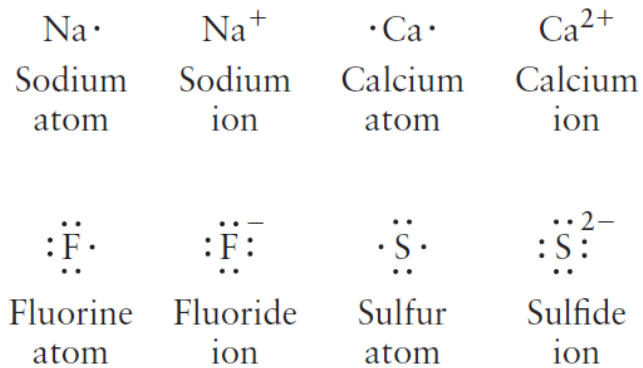
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Ionic bonding (离子键)

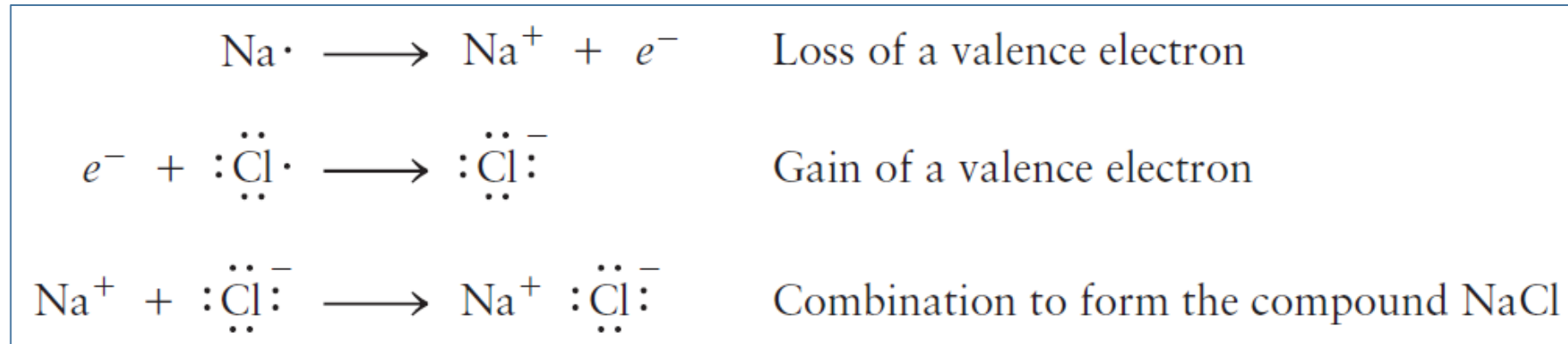
An **ionic bond** forms between atoms with large differences in electronegativity, such as Na and F.



The more electropositive atom **transfers an electron** to the more electronegative atom, forming an ion pair that is then drawn together by the attractive electrostatic force.



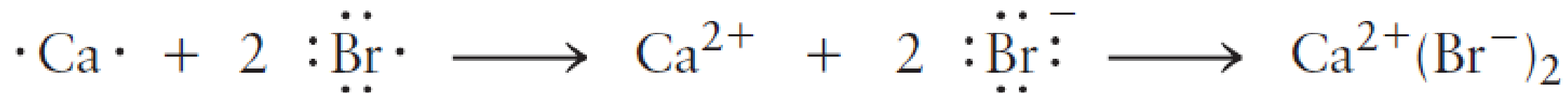
Lewis symbols



ionic compound

Ionic bonding

The **total charge** on the compound = **0**, thus the stoichiometry is determined by charge balance.



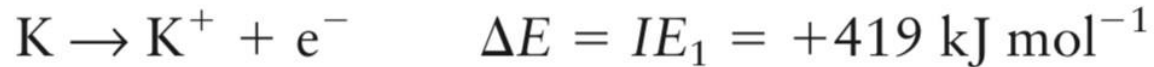
Neutral atoms
not having octets

Positive ion
with octet

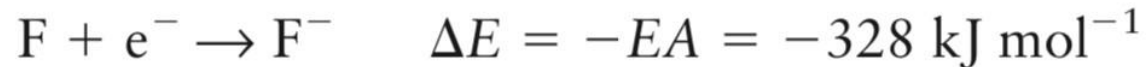
Negative ions
with octets

Ionic
compound

Ionic bonding



$$\Delta E_\infty = IE_1(\text{K}) - EA(\text{F}) = +91 \text{ kJ mol}^{-1}$$



The total energy cost for the creation of this ion pair is always **positive**:

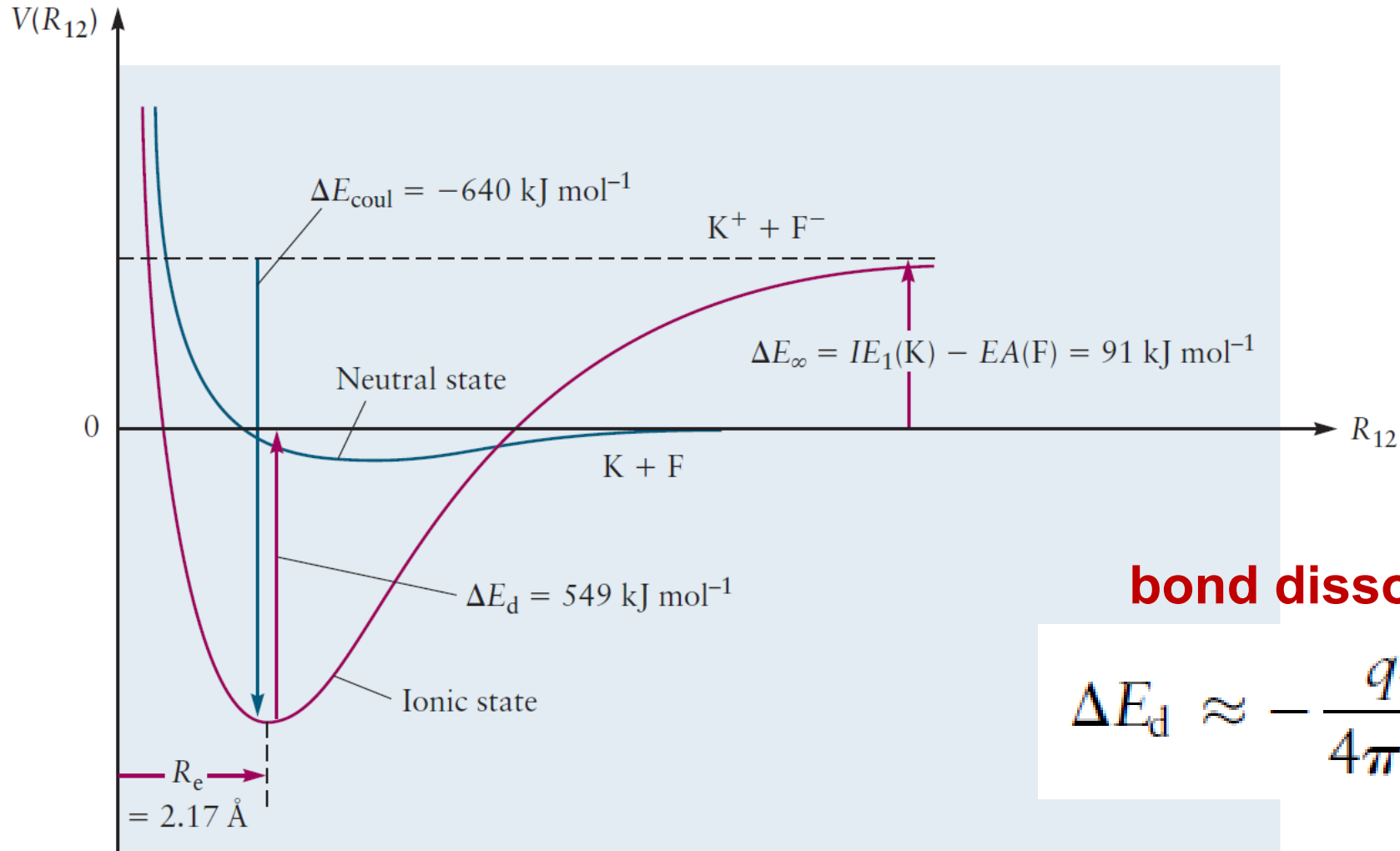
the smallest ionization energy of any element > the largest electron affinity of any element.

$$V(R_{12}) = \frac{q_1 q_2}{4\pi\epsilon_0 R_{12}} \text{ (J per ion pair)}$$

$$V(R_{12}) = \frac{q_1 q_2}{4\pi\epsilon_0 R_{12}} \cdot \frac{N_A}{10^3} \text{ (kJ mol}^{-1}\text{)}$$

The ions are attracted to one another by the electrostatic force, which is described by Coulomb's law. This is called **Coulomb stabilization energy**.

Ionic bonding



bond dissociation energy

$$\Delta E_d \approx -\frac{q_1 q_2}{4\pi\epsilon_0 R_e} \frac{N_A}{10^3} - \Delta E_{\infty}$$

FIGURE 3.13 The potential energy of the ions K^+ and F^- , and the atoms K and F , as a function of their internuclear separation R_{12} .

Question

Estimate the energy of dissociation to neutral atoms for KF, which has a bond length of 2.17×10^{-10} m. For KF, $\Delta E_{\infty} = IE_1(\text{K}) - EA(\text{F}) = 91 \text{ kJ mol}^{-1}$.

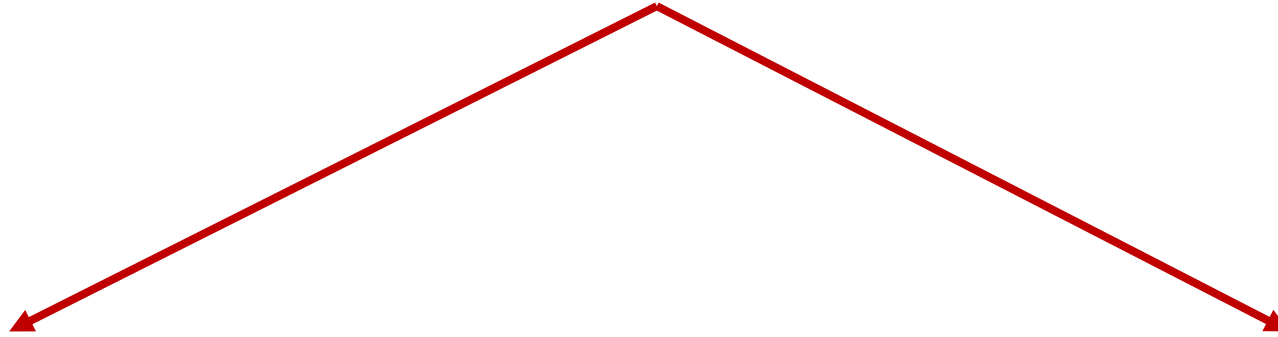
SOLUTION

$$\begin{aligned}\Delta E_{\text{d}} &\approx -\frac{q_1 q_2}{4\pi\epsilon_0 R_{\text{e}}} \frac{N_{\text{A}}}{10^3} - \Delta E_{\infty} \\&= -\frac{-(1.602 \times 10^{-19} \text{ C})^2 (6.022 \times 10^{23} \text{ mol}^{-1})}{(4)(3.1416)(8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1})(2.17 \times 10^{-10} \text{ m})(10^3 \text{ J kJ}^{-1})} \\&\quad - 91 \text{ kJ mol}^{-1} \\&= 640 \text{ kJ mol}^{-1} - 91 \text{ kJ mol}^{-1} \\&= 549 \text{ kJ mol}^{-1}\end{aligned}$$

This estimate compares fairly well with the experimentally measured dissociation energy of 498 kJ mol^{-1} .

Ionic bonding

Why this simple model does not do a better job in calculating the bond energy?



All bonds have some degree of **covalent character**, which involves **electron sharing** between the atoms.

We have assumed that each ion is a point charge. In reality, the distribution of electrons distorted by presence of the potassium ion; this distortion is called **polarization**.

Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**
 1. Forces and potential energy in molecules: formation of chemical bonds
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 3. Covalent and polar covalent bonding
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Covalent bonding (共价键)

What kinds of bonds are formed between elements of identical or comparable electronegativities (e.g. H₂ or CO)?



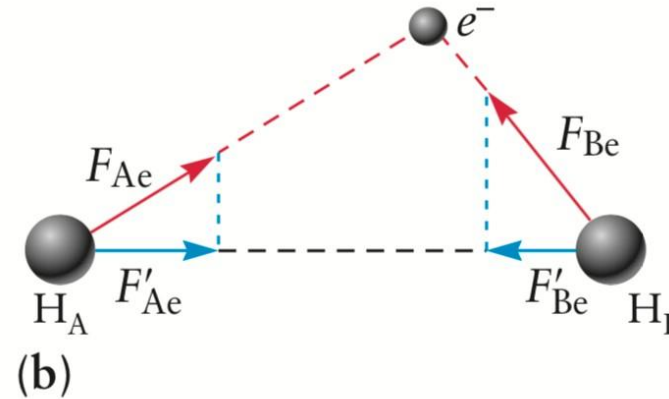
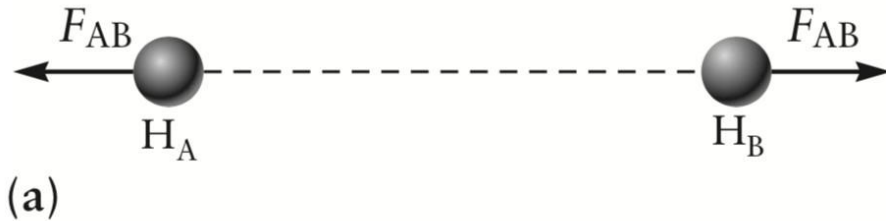
No Coulomb stabilization energy from ions with opposite charges



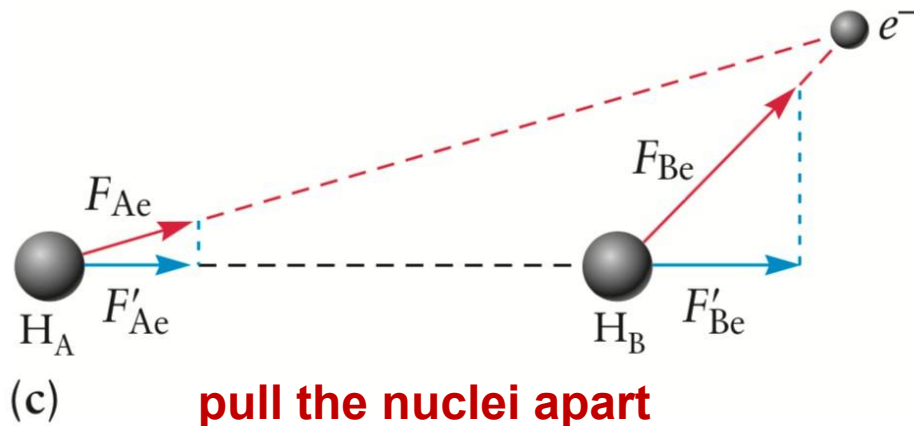
What is the **driving force** for bond formation from separated atoms in the gas phase?

Covalent bonding

A plausibility argument to suggest why a covalent bond might form, by focusing on the forces acting on the nuclei due to the electrons



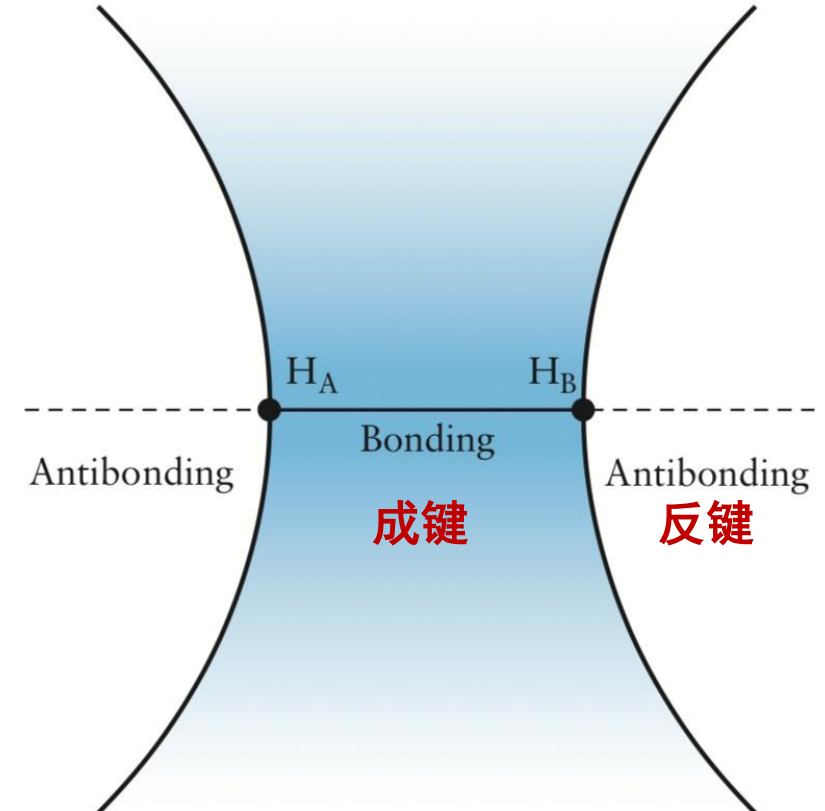
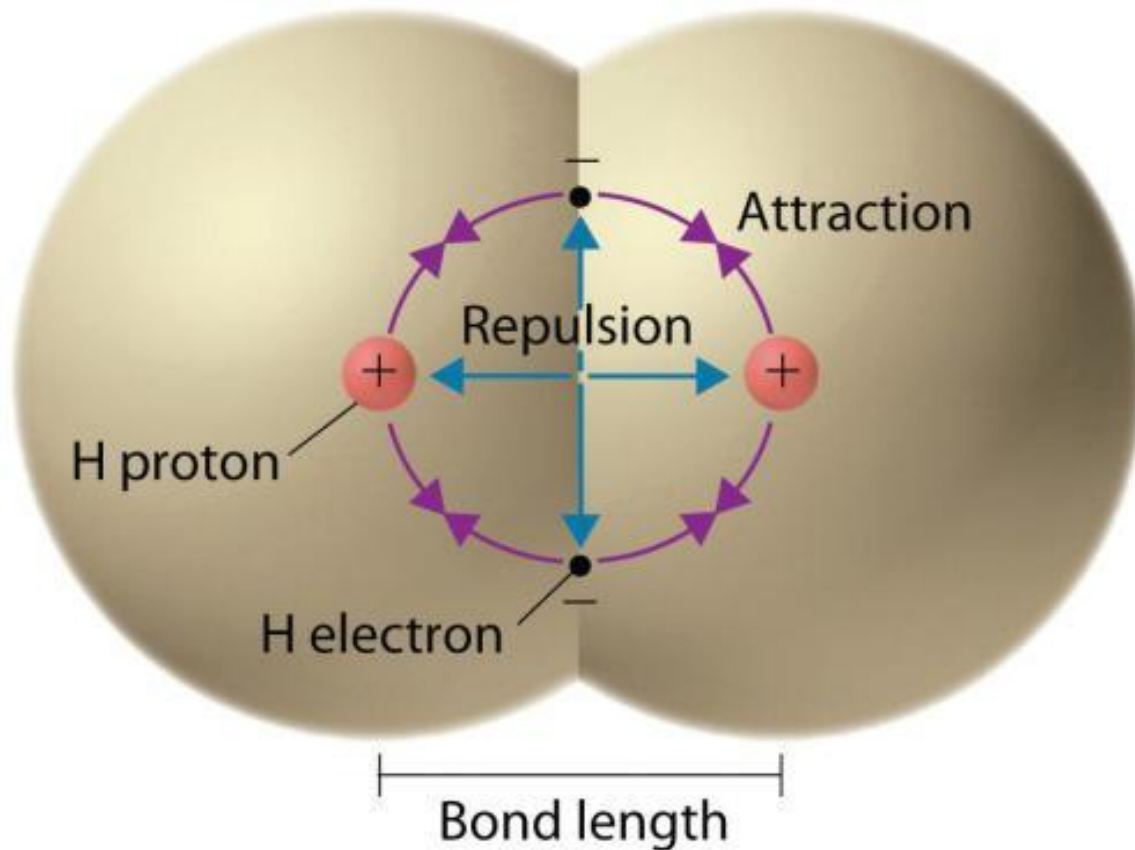
pull the nuclei together



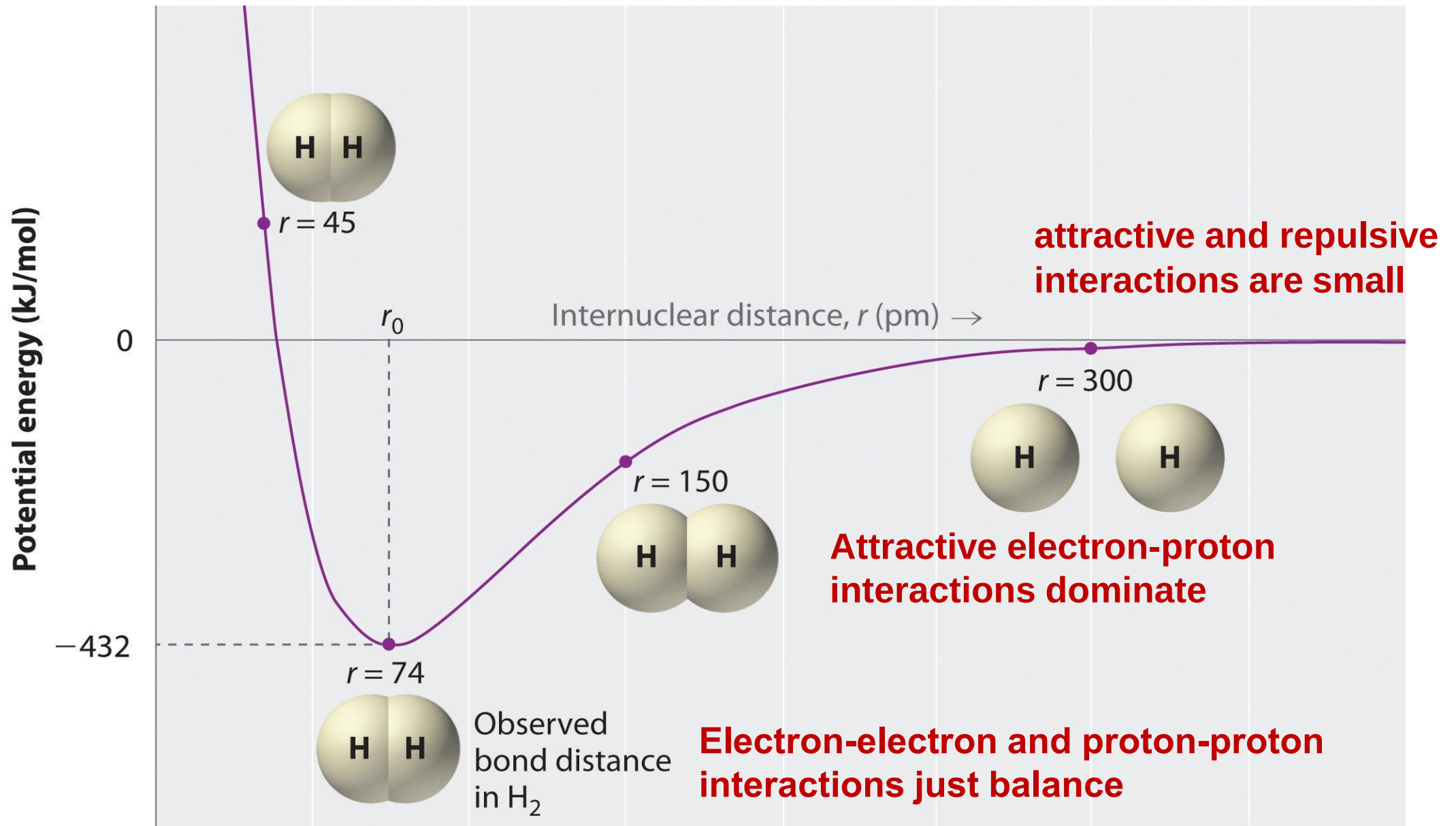
pull the nuclei apart

Covalent bonding

- This simple model is supported by experimental data for H_2^+ .
- $R_e = 1.06 \text{ \AA}$, $E_d = 255.5 \text{ KJ mol}^{-1}$, which is characteristic of a stable covalent bond.



Covalent bonding



Covalent and polar covalent bonding

- Bond length (键长 R_e):** the distance between the nuclei of the two atoms

Molecule	Bond Length (Å)	Bond Energy (kJ mol ⁻¹)
H ₂	0.751	433
N ₂	1.100	942
O ₂	1.211	495
F ₂	1.417	155
Cl ₂	1.991	240
Br ₂	2.286	190
I ₂	2.669	148
HF	0.926	565
HCl	1.284	429
HBr	1.424	363
HI	1.620	295
ClF	1.632	252
BrF	1.759	282
BrCl	2.139	216
ICl	2.324	208
NO	1.154	629
CO	1.131	1073

Within the same group, bond length increases with increasing atomic number

TABLE 3.4 Reproducibility of Bond Lengths

Bond	Molecule	Bond Length (Å)
O—H	H ₂ O	0.958
	H ₂ O ₂	0.960
	HCOOH	0.95
	CH ₃ OH	0.956
C—C	Diamond	1.5445
	C ₂ H ₆	1.536
	CH ₃ CHF ₂	1.540
	CH ₃ CHO	1.50

Length of a bond of the same type between a given pair of atoms changes little

Covalent and polar covalent bonding

- Bond energy (键能 ΔE_d):** the energy required to break one mole of the particular bond.

Molecule	Bond Length (Å)	Bond Energy (kJ mol ⁻¹)
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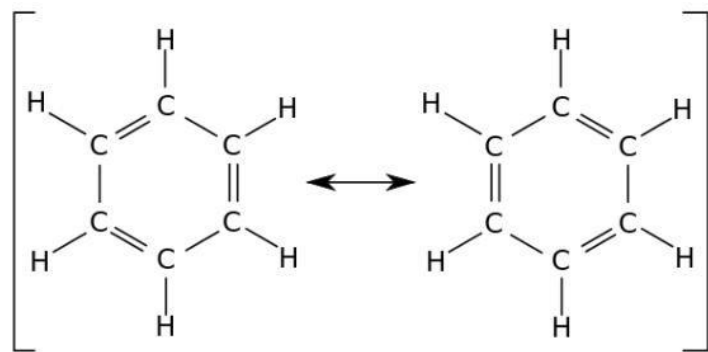
HF>HCl>HBr>HI

Covalent bonding

- **Bond order (键级):** the number of shared electron pairs between the two atoms.
 - 键级是衡量化学键相对强弱的参数，键级愈大，键愈稳定
 - 若键级为零，则不能成键。

TABLE 3.5 Three Types of Carbon–Carbon Bonds

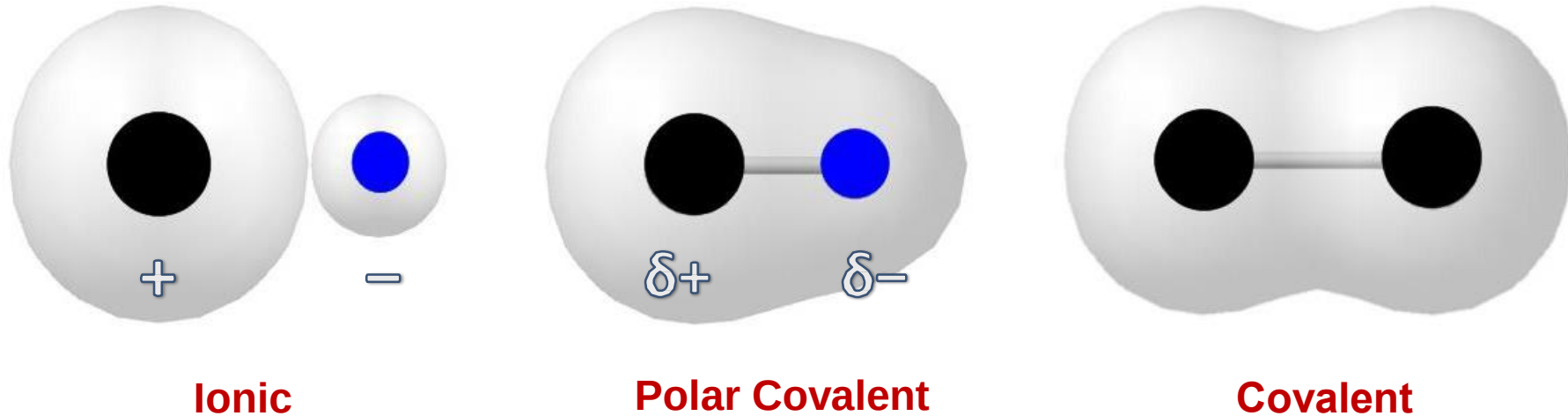
Bond	Molecule	Bond Length (Å)	Bond Energy (kJ mol ⁻¹)
C—C	C ₂ H ₆ (or H ₃ CCH ₃)	1.536	345
C=C	C ₂ H ₄ (or H ₂ CCH ₂)	1.337	612
C≡C	C ₂ H ₂ (or HCCH)	1.204	809



1.397 Å

505 kJ mol⁻¹

Polar covalent bonding (极性共价键)



- **Polar Covalent:** bonds in which there is a partial transfer of charge
- The absolute value of the **difference in electronegativity** of two bonded atoms tells the **degree of polarity** in their bond.

Ionic	Polar Covalent	Covalent
$\Delta\chi > 2.0$		$\Delta\chi < 0.4$

Dipole moments and percent ionic character

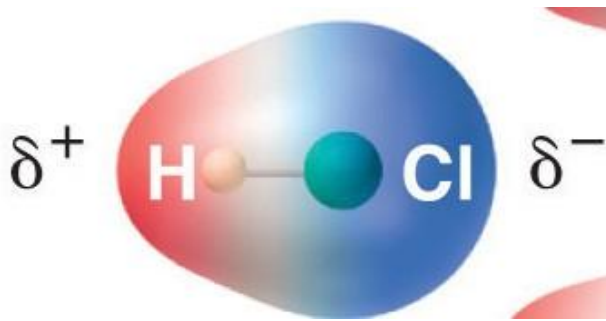
If two charges of equal magnitude and opposite sign, $+q$ and $-q$, are separated by a distance R , the **dipole moment μ (偶极矩)** of that charge distribution is:

$$\mu = qR$$

debye (D), $1\text{D} = 3.336 \times 10^{-30} \text{ C m}$



The debye can also be defined as the dipole moment of two charges $\pm e$ separated by $0.2082 \text{ \AA}$.



$$\mu(\text{D}) = [R(\text{\AA})/0.2082 \text{ \AA}] \delta$$

If δ is the fraction of a unit charge on each atom in a diatomic molecule ($q = e\delta$) and R is the equilibrium bond length

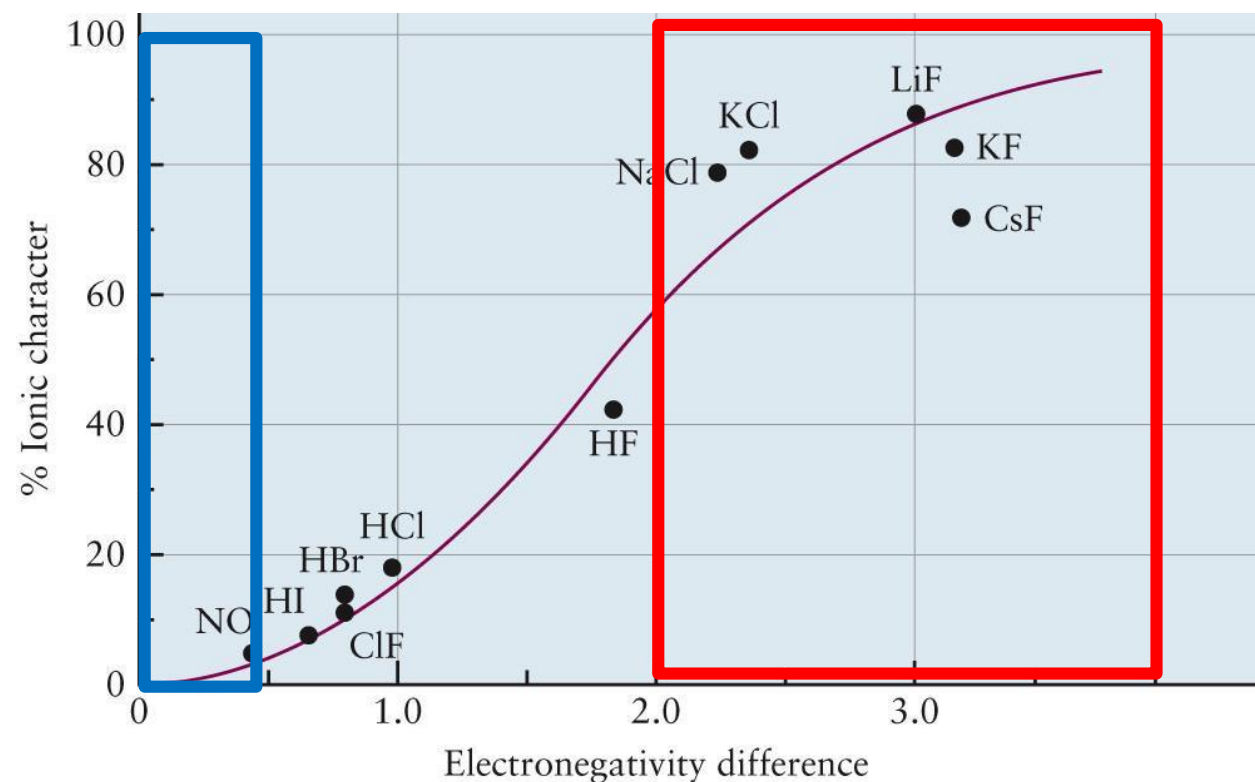
δ is the fraction ionic character

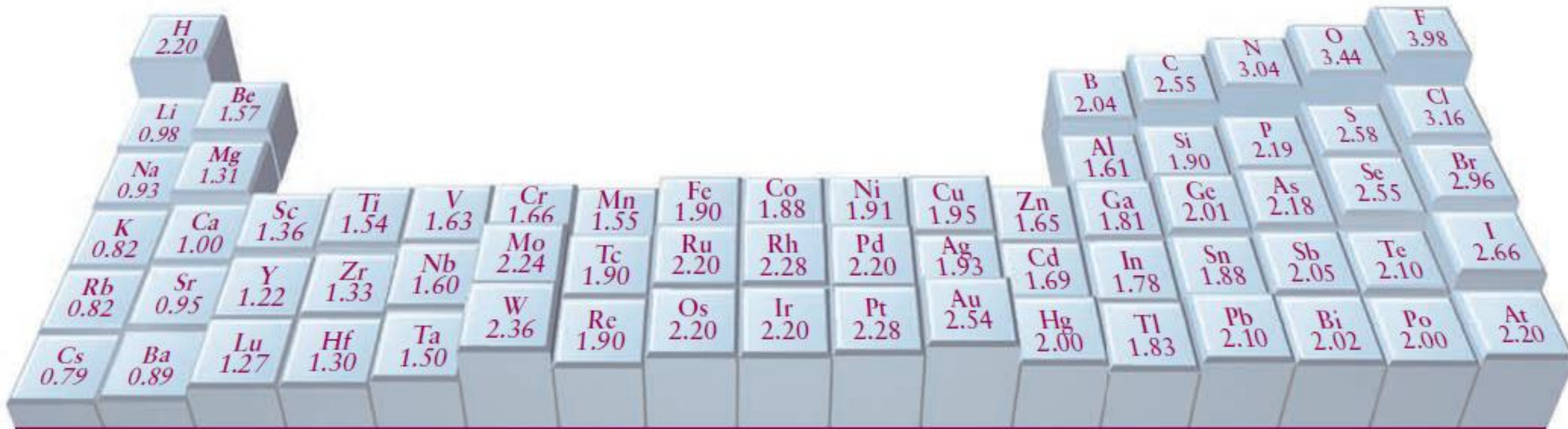
Dipole moments and percent ionic character

TABLE 3.7

Dipole Moments of Diatomic Molecules

Molecule	Bond Length (Å)	Dipole Moment (D)	% Ionic Character (100 δ)
H ₂	0.751	0	0
CO	1.131	0.112	2
NO	1.154	0.159	3
HI	1.620	0.448	6
ClF	1.632	0.888	11
HBr	1.424	0.828	12
HCl	1.284	1.109	18
HF	0.926	1.827	41
CsF	2.347	7.884	70
LiCl	2.027	7.129	73
LiH	1.604	5.882	76
KBr	2.824	10.628	78
NaCl	2.365	9.001	79
KCl	2.671	10.269	82
KF	2.176	8.593	82
LiF	1.570	6.327	84
NaF	1.931	8.156	88





Use above figure, arrange the following bonds in order of decreasing polarity: H-C, O-O, H-F, I-Cl, Cs-Au.

Question

Using Figure 3.10, arrange the following bonds in order of decreasing polarity: H—C, O—O, H—F, I—Cl, Cs—Au.

SOLUTION

The differences in electronegativity among the five pairs of atoms (without regard to sign) are 0.35, 0.00, 1.78, 0.50, and 1.75, respectively. The order of decreasing polarity is the same as the order of decrease in this difference: H—F, Cs—Au, I—Cl, H—C, and O—O. The last bond in this listing is nonpolar.

Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**
 1. Forces and potential energy in molecules: formation of chemical bonds
 2. Ionic bonding
 3. Covalent and polar covalent bonding
 4. Electron pair bonds and Lewis diagrams for molecules
 5. The shapes of molecules
 6. Oxidation numbers
 7. Inorganic nomenclature

Lewis model

Lewis dot symbol

- The valence electrons as dots arranged around the chemical symbol
- The core electrons are not shown.
- The first four dots are arranged individually around the four sides of the symbol.
- If an atom has more than four valence electrons, dots representing them are then paired with those already present.



Gilbert N. Lewis
(Berkeley, 1875–1946)



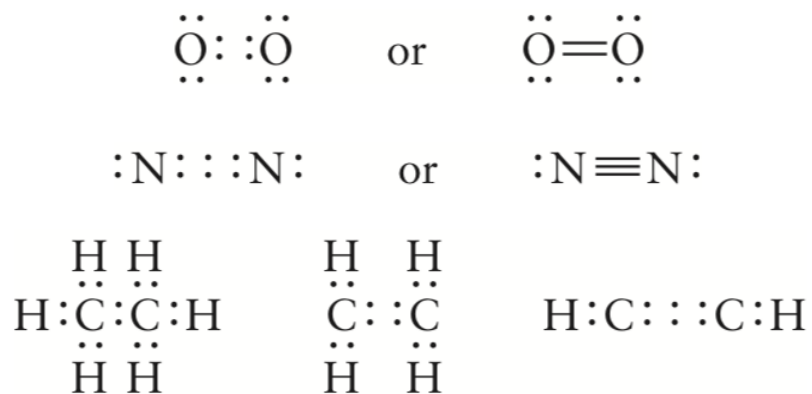
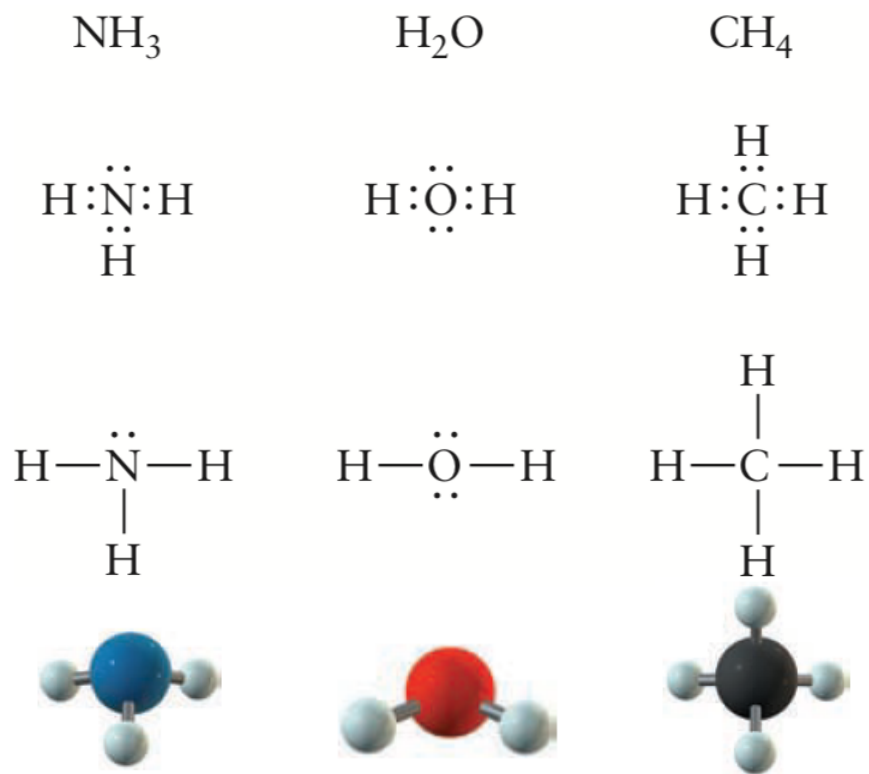
Lewis, G. N. "The Atom and the Molecule", *J. Am. Chem. Soc.* **1916**, 38, 762–785.

<https://www.youtube.com/watch?v=cluXl7o6mAw&spfreload=1>

Lewis diagrams

Octet rule (八隅律) : Whenever possible, the electrons in a covalent compound are distributed in such a way that each main-group element (except H) is surrounded by eight electrons.

- A **shared pair electrons** = a **covalent bond**
- A **lone pair electrons** = **no bonding**
- Each atom achieves its own **noble-gas shell** of electrons
- more than one pair may be shared (**double bond, triple bond**)



Lewis diagrams: formal charges

Formal charge (形式电荷): the charge an atom in a molecule would have if the electrons in its Lewis diagram were divided equally among the atoms that share them.



Formal charge = number of valence electrons

- number of electrons in lone pairs**
- $\frac{1}{2}$ (number of electrons in bonding pairs)**

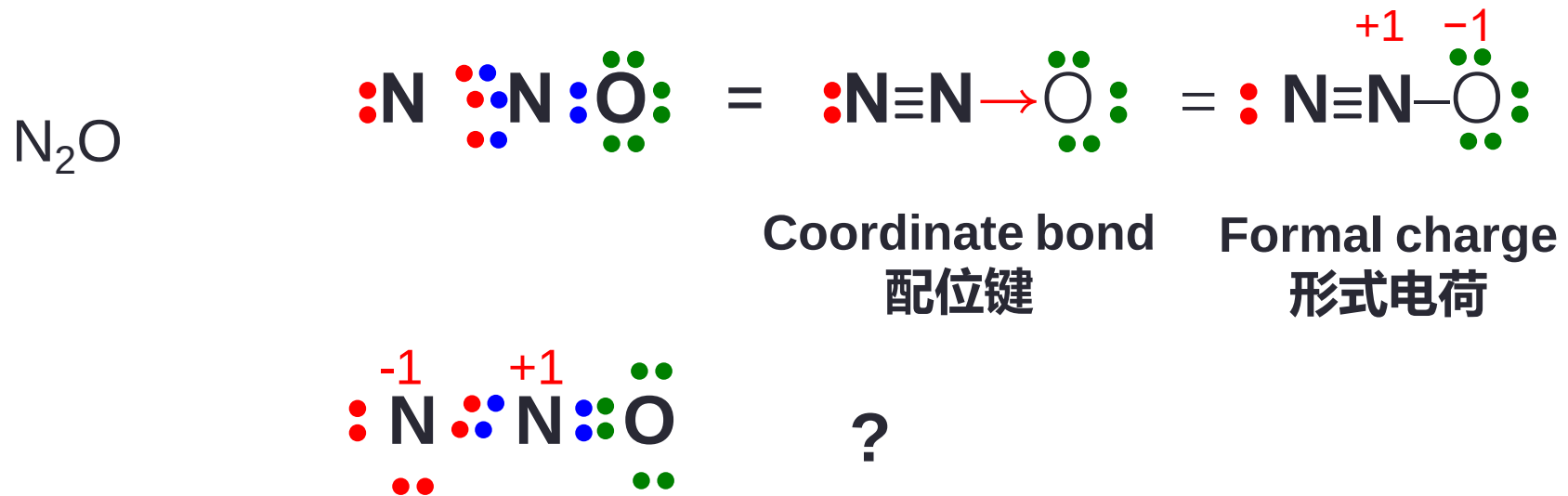
Lewis diagrams: formal charges and coordinate bonds



Formal charge (empirical speculation):

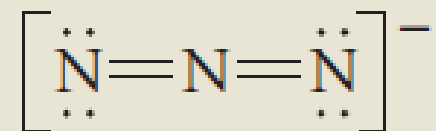
1. The structure should have less charge.
2. Same kind of charge will not be located on adjacent atoms.
3. Minus charge tend to be on the atom with higher negativity.

Application of Lewis dot structure



Lewis diagrams: formal charges

Compute the formal charges on the atoms in the following Lewis diagram, which represents the azide ion (N_3^-).



Solution

Nitrogen is in Group V. Hence, each N atom contributes 5 valence electrons to the bonding, and the negative charge on the ion contributes 1 more electron. The Lewis diagram correctly represents 16 electrons.

Each of the terminal nitrogen atoms has four electrons in lone pairs and four bonding electrons (which comprise a double bond) associated with it. Therefore,

$$\text{formal charge}_{(\text{terminal N})} = 5 - 4 - \frac{1}{2}(4) = -1$$

The nitrogen atom in the center of the structure has no electrons in lone pairs. Its entire octet comprises the eight bonding electrons:

$$\text{formal charge}_{(\text{central N})} = 5 - 0 - \frac{1}{2}(8) = +1$$

The sum of the three formal charges is -1 , which is the true overall charge on this polyatomic ion. Failure of this check indicates an error either in the Lewis diagram or in the arithmetic.

Write a Lewis electron dot diagram for phosphoryl chloride, POCl_3 (Fig. 3.18). Assign formal charges to all the atoms.

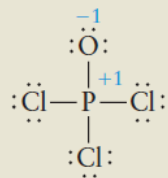
Solution

The first step is to calculate the total number of valence electrons available in the molecule. For POCl_3 , it is

$$5 \text{ (from P)} + 6 \text{ (from O)} + [3 \times 7 \text{ (from Cl)}] = 32$$

Next, calculate how many electrons would be necessary if each atom were to have its own noble-gas shell of electrons around it. Because there are 5 atoms in the present case (none of them hydrogen), 40 electrons would be required. From the difference of these numbers ($40 - 32 = 8$), each atom can achieve an octet only if 8 electrons are shared between pairs of atoms. The 8 electrons correspond to 4 electron pairs, so each of the four linkages in POCl_3 must be a single bond. (If the number of shared electron pairs were *larger* than the number of bonds, double or triple bonds would be present.)

The other 24 valence electrons are assigned as lone pairs to the atoms in such a way that each achieves an octet configuration. The resulting Lewis diagram is



Formal charges are already indicated in this diagram. Phosphorus has the group number V, and it shares eight electrons with no lone-pair electrons, so

$$\text{formal charge on P} = 5 - 4 = +1$$

Oxygen has the group number VI with six lone-pair electrons and two shared electrons, so

$$\text{formal charge on O} = 6 - 6 - \frac{1}{2}(2) = -1$$

All three chlorine atoms have zero formal charge, computed by

$$\text{formal charge on Cl} = 7 - 6 - \frac{1}{2}(2) = 0$$

How to draw Lewis diagrams

Step 1: calculate total number of valence electrons + negative ion - positive ion



$$5 + 6 + 3 \times 7 = 32$$

Step 2: calculate number of electrons needed to form noble-gas shell

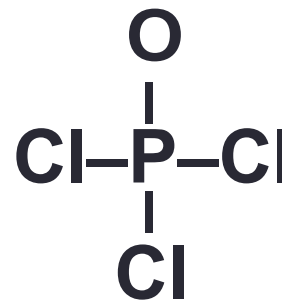
$$5 \times 8 = 40$$

Step 3: calculate numbers of shared electrons = step 2 - step 1

$$40 - 32 = 8$$

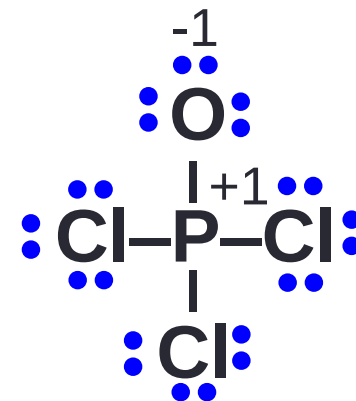
Step 4: assign two bonding electrons to each bond

Step 5: assign remaining bonding electrons to make double or triple bonds



Step 6: assign remaining electrons as lone pairs to the atoms, giving octets to all atoms except H.

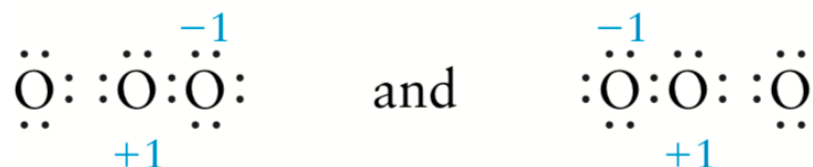
Step 7: Determine the formal charge on each atom and write it next to that atom.



Step 8: If more than one diagram is possible, choose the one with the smallest magnitudes of formal charges and with any negative formal charges placed on the most electronegative atoms

Lewis diagrams: resonance forms

For certain molecules or molecular ions, two or more **equivalent Lewis diagrams** can be written



Are there any difference between two O-O and O=O bonds?

The two O-O bond lengths are found experimentally to be identical (1.28 Å), which is intermediate between the O-O single bond length in H₂O₂ (1.49 Å) and the O=O double bond length in O₂ (1.21 Å).



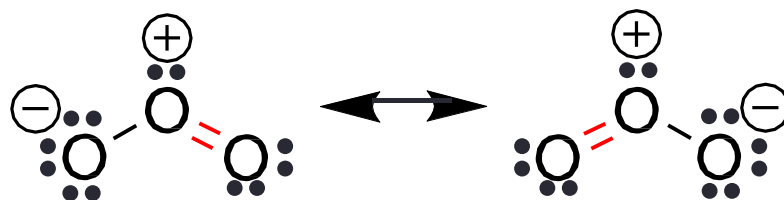
resonance hybrid



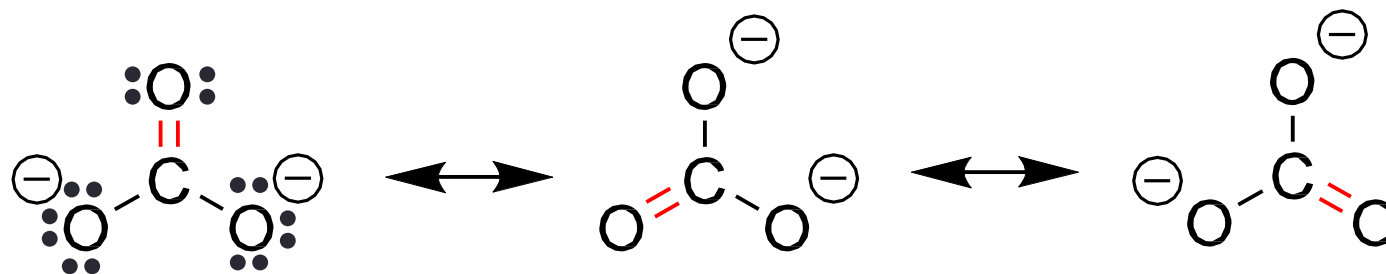
The term **resonance** (共振) does not mean that the molecule physically oscillates. The best representation of the actual bonding is a hybrid diagram that includes features of each of the acceptable individual diagrams.

Lewis diagrams: resonance forms and Bond Order

$$\text{B.O. (Resonance hybrid)} \approx \overline{\text{B.O. (Canonical forms)}}$$



O–O bond order = 1.5



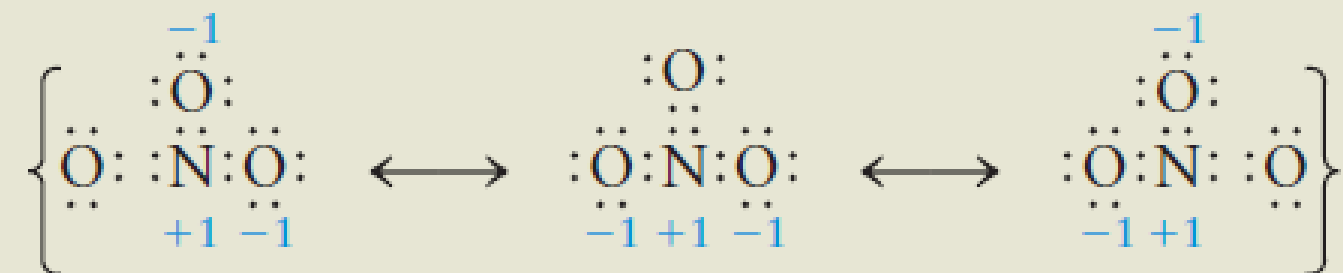
C–O bond order $\approx$ 1.3



Draw three resonance forms for the nitrate ion NO_3^- , and estimate the bond lengths.

Solution

NO_3^- contains 24 valence electrons. For each atom to have its own octet, $4 \times 8 = 32$ electrons would be required. Therefore, $32 - 24 = 8$ electrons must be shared between atoms, implying a total of four bonding pairs. These can be distributed in one double and two single bonds, leading to the equivalent resonance diagrams:



The two singly bonded oxygen atoms carry formal charges of -1 , and the charge of the nitrogen atom is $+1$. The bond lengths should all be equal and should lie between the values given in Table 3.5 for $\text{N}-\text{O}$ ($1.43 \text{ \AA}$) and $\text{N}=\text{O}$ ($1.18 \text{ \AA}$). The experimentally measured value is $1.24 \text{ \AA}$.

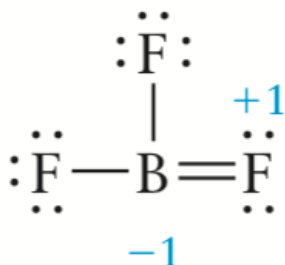
Lewis diagrams: breakdown of the octet rule

Case 1: odd-electron molecules

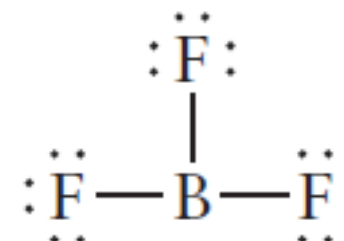


Nitrogen oxide has 11 electrons, and this is the best electron dot diagram.

Case 2: octet-deficient molecules



Experimental evidence suggests
there are no double bonds in BF_3

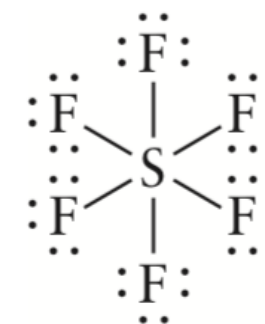


Case 3: valance shell expansion

$$56(7 \times 8) - 48(6 + 6 \times 7) = 8 \text{ bonding electrons}$$



Step 6': abandon the octet rule for the central atom but preserve it for the terminal atoms. If any electrons still remain, assign them to the central atoms as lone pairs if the central atoms is not one from the 2nd period

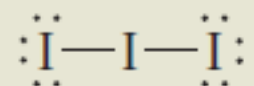


Write a Lewis diagram for I_3^- , the linear (tri-iodide) ion.

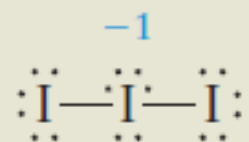
Solution

There are 7 valence electrons from each iodine atom plus 1 from the overall ion charge, giving a total of 22. Because $3 \times 8 = 24$ electrons would be needed in separate octets, $24 - 22 = 2$ are shared according to the original rules. Two electrons are not sufficient to make two different bonds, however, so valence expansion is necessary.

A pair of electrons is placed in each of the two bonds, and rule 6' is used to complete the octets of the two terminal I atoms. This leaves



At this stage, 16 valence electrons have been used. The remaining 6 are placed as lone pairs on the central I atom. A formal charge of -1 then resides on this atom:

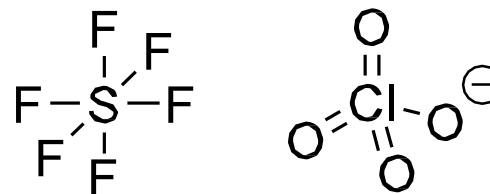


Note the valence expansion on the central atom: It shares or owns a total of 10 electrons, rather than the 8 required by adherence to the octet rule.

Summary

Molecular Orbital	Lewis Dot Structure
Bonding orbital	Shared electron pair
Nonbonding orbital	Lone electron pair
Delocalized orbital	Resonance structures
Two electrons from one atom	Coordinate bond
	Formal charge

- The **octet rule** works best for **C, N, O, F**.



Homework: read “principle of modern chemistry”, chapter 3